

Pornhibition: Analyzing the Effects of Age Verification Laws on Pornography Consumption and Risky Behaviors

Emily Davis*

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Abstract

The top three pornography websites receive nearly 8 billion monthly visits globally—more than Amazon, TikTok, OpenAI, and Netflix combined (Wright, Tounaga, & Herbenick, 2023). A large body of research links adolescent pornography consumption to increased sexual aggression, risky sexual behavior, and worsening mental health. In response to growing concerns about the health and safety of minors, Louisiana became the first state to enact an age verification law on January 1, 2023. This law requires websites with at least 33.3 percent pornographic content to verify that users are at least 18 years old, or else face substantial fines. Since then, 24 additional states have passed similar age verification laws. In this paper, I evaluate the impact of age verification laws on pornography consumption using Google Trends data. Using an event study design that exploits the staggered adoption of these laws, I find that search interest in the most popular pornographic site plummets after age verification laws go into effect, persisting for the entire post-period of six months. However, search interest in the two leading non-compliant sites spike after age verification laws go into effect, remaining significantly elevated for the entire post-period. I also find evidence of increased search interest in VPNs that is significant for up to four months after age verification laws go into effect. My findings suggest that the effectiveness of age verification laws may be undermined by the availability of substitutes and technological workarounds.

*University of Notre Dame Department of Economics

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1 Introduction

The top three pornography websites receive nearly 8 billion monthly visits globally—more than Amazon, TikTok, OpenAI, and Netflix combined (Wright et al., 2023). Though pornography is created for adult consumption, in the age of the internet, it is increasingly easy for minors to access this content. In the U.S. alone, more than half of children have viewed pornography before the age of 13 (Robb and Mann, 2023). A large body of research links adolescent pornography consumption to increased sexual aggression, risky sexual behavior, and worsening mental health. In response to growing concerns about the health and safety of children, Louisiana became the first state to enact an age verification law on January 1, 2023. This law requires websites with a substantial portion of pornographic content to verify that users are at least 18 years old, or else face substantial fines. Since then, 24 additional states have passed similar age verification laws.

Despite the growing adoption of age verification laws, there is little empirical evidence on their effectiveness in limiting minors’ access to pornography and their impact on pornography consumption and risky behaviors in the broader population. On one hand, by requiring users to provide personal information to pornography sites, these laws could reduce consumption by creating a significant privacy barrier. This could deter not only minors but also adults from visiting pornography sites. On the other hand, users may instead turn to non-compliant sites or employ Virtual Private Networks (VPNs) to bypass restrictions, leading to little or no change in overall consumption. By seeking out non-compliant sites, users may also risk encountering more harmful content and less secure management of their personal data.

In this paper, I evaluate the impact of age verification laws on pornography consumption using Google Trends data. Using an event study design that exploits the staggered adoption of these laws, I find that search interest in the most popular pornographic site (Pornhub) plummets after age verification laws go into effect, persisting for the entire post-period of six months. However, search interest in the two leading non-compliant sites (XNXX and Xvideos) spike after age verification laws go into effect, remaining significantly elevated for the entire post-period. I also find evidence of increased search interest in VPNs that is significant for up to four months after age verification laws go into effect. My findings suggest that the effectiveness of age verification laws may be undermined by the availability of substitutes and technological workarounds.

My paper is related to a very recent working paper by Lang et al. (2025) which uses data from Google Trends in a synthetic control design to measure how search interest in Pornhub, Xvideos, VPNs, and adult content more generally evolved in response to age verification laws. My results complement the findings of this paper by demonstrating the robustness of the effects to different empirical specifications. I also expand the analysis to analyze the effects of age verification laws on a compliant site, Xhamster, and an additional non-compliant site, XNXX. Pornhub chose to block access to its site in the vast majority of states that have enacted age verification laws. Because of this, the decline in search interest for Pornhub could reflect a mechanical drop in interest due to users not being able to access

the site. Xhamster has instead chosen to comply with the law, requiring users to verify their age before accessing content. Thus, my results for Xhamster provide insight into the behavior of those who cannot verify their age or those who do not wish to verify their age by providing personally identifying information. By expanding the analysis to include an additional non-compliant site, I demonstrate that there are potentially many substitutes for blocked or compliant sites.

My paper is also related to [Madio et al. \(2025\)](#), who examine how asymmetric content moderation of adult websites impacts traffic to unregulated sites and competition in search. In December of 2020, the New York Times published an opinion piece titled “The Children of Pornhub,” which exposes how Pornhub profited from non-consensual content, especially content featuring the exploitation of minors ([Kristof, 2020](#)). MindGeek, the parent company of Pornhub and other adult websites, responded to these allegations by removing all unverified videos from its platform, which amounted to deleting 80 percent of Pornhub’s video library. The authors’ estimates imply that the traffic MindGeek sites lost after this purge was diverted entirely to competing firms—primarily fringe websites. This finding is relevant to the context in my paper because it demonstrates that the market for pornography is highly competitive, enabling consumers to substitute almost perfectly between websites. The authors also propose that their findings are an illustration of the *balloon effect*: when regulation is applied to only one part of a system, the negative consequences are not eliminated, but displaced to parts with lower resistance. Age verification laws only apply to pornography sites, despite the widespread presence of pornography on social media sites.¹ Even among pornography sites, many choose not to comply. Thus, the findings of this paper further motivate the importance of understanding the unintended consequences of age verification laws.

The rest of my paper proceeds as follows. In [Section 2](#) I provide background information on age verification laws, describing what these laws entail and how pornography sites and the legal system have responded to them. In [Section 3](#) I describe the Google Trends data I use in my analysis. In [Section 4](#) I present descriptive evidence for the impact of age verification laws on pornography consumption. In [Section 5](#), I describe the event study analysis I use to estimate the causal effect. In [Section 6](#), I discuss the results, largely formalizing the takeaways from the descriptive evidence in the earlier section. I conclude with a discussion of my results and future steps in [Section 7](#).

¹[Robb and Mann \(2023\)](#) report that among teens who intentionally consumed pornography (44 percent of all teens in their sample), 38 percent of them accessed pornography at least weekly on social media sites such as TikTok, Instagram, Snapchat, and Reddit. Social media was the second most popular destination for pornography. Porn websites such as Pornhub, YouPorn, and Xvideos were the most popular destination, with 44 percent of teens visiting them at least weekly.

2 Background

On January 1, 2023, Louisiana’s House Bill 142 (HB 142) went into effect, making it the first ever age verification law for pornography sites in the United States. Since then, 24 additional states have passed similar age verification laws. By the end of 2025, half of all US states will have age verification laws in effect. Figure 2 and Table 1 highlight the widespread adoption of these laws and the variation in the timing of their passage and effective dates.

Louisiana’s law established the legal template for the age verification laws passed in the subsequent states: the vast majority have passed copycat laws with few deviations from the original. In the following subsections, I describe the content of these laws and the response these laws have garnered from pornography sites themselves as well as the court system.

2.1 What are Age Verification Laws?

According to the language from Louisiana’s HB 142, age verification laws require “commercial entities”—who distribute “material harmful to minors” via websites that contain a “substantial proportion” of such material—to perform “reasonable age verification methods.” The law also establishes that any commercial entity that fails to perform age verification methods is “liable to an individual for damages.” To give the law more bite, Louisiana enacted a follow up law—HB 77—a few months later in August, 2023. HB 77 establishes civil penalties for violating the law and states that commercial entities are not only liable to individuals for violations, but the state attorney general as well.²

The states following in Louisiana’s footsteps have largely adopted the language and features of Louisiana’s laws. In what follows I describe the terminology quoted above in more detail and what this means in practice for those affected by these laws. Where relevant, I denote any important deviations other states have made from the original Louisiana laws.³

The term “commercial entities” refers to corporations, limited liability companies, partnerships, and other legally recognized entities that operate the websites in question. Importantly, many age verification laws explicitly state that internet service providers, search engines, and cloud servers are not liable for any role they play in facilitating access to pornographic material. These laws target operators of websites directly.

The phrasing “substantial proportion” of “material harmful to minors” determines the types of websites targeted by the laws. These definitions ensure that the laws apply specif-

²The civil penalties described in HB 77 can reach up to \$5,000 for each day of violation. The attorney general may also request that the court impose an additional \$10,000 for each section of the law the commercial entity violates. Besides these civil penalties, commercial entities are also responsible for all costs, expenses, and fees related to the investigation, including attorney fees.

³For the specific language of each state’s law, please consult the full text of each law on the state legislatures’ websites using the names in Table 1 as a reference.

ically to pornography sites, not just any site on which pornography may appear.⁴ Material harmful to minors is typically defined as “any material that the average person, applying contemporary community standards would find, taking the material as a whole and with respect to minors, is designed to appeal to, or is designed to pander to, the prurient interest.”⁵ The laws carve out an exception for material that has serious literary, artistic, political, or scientific value for minors. In the vast majority of states “substantial proportion” refers to sites in which one third or more of the content is pornographic.⁶ In practice, the purpose of specifying an exact percentage seems to be to distinguish sites that primary function as hosts for pornographic material from sites in which pornographic material comprises an incidental share of the content. Thus far, this technical definition has not been salient for legal challenges to these laws or how the pornography sites have responded to them.⁷

The age verification aspect of these laws requires websites to verify the age of visitors either directly or through a third-party age verification service using “reasonable age verification methods.” Common “reasonable age verification methods” include the following: uploading a government-issued ID, such as a drivers’ license or passport; providing credit card information; or completing a biometric verification method, such as facial recognition or voice analysis, in which artificial intelligence is employed to estimate users’ age. Many websites choose to partner with third-party age verification services instead of implementing the age verification method themselves. This is likely because most states have stipulations for how user data is to be handled, specifying substantial civil penalties for the misuse of user data.

Finally, these laws are enforced through the court system: individuals or state attorneys general can sue commercial entities that fail to perform adequate age verification methods for damages. These damages could include the consequences resulting from a minor’s accessing harmful material, or from the misuse of an individual’s personal data. The laws also hold the commercial entities responsible for any expenses incurred in the investigation process. The enforcement aspect of the law is where there is the greatest range of differences among states. States differ in who can bring forth the case (individuals or attorneys general) and the size of the civil penalties for violating the law.⁸

⁴Social media is the second most popular source for viewing pornography among teens. [Robb and Mann \(2023\)](#) report that among teens who intentionally consume pornography (44% of all teens), 38% consume pornography weekly on social media, while 44% consume it weekly on pornography sites.

⁵This definition of material harmful to minors is taken from *Ginsberg v. New York* (1968).

⁶Notable exceptions are Kansas, which lowers the threshold to 25 percent pornographic content, and Missouri, South Dakota, and Wyoming have no limit on the percentage of harmful content ([Coalition, 2025](#)).

⁷For example, I am not aware of evidence showing that pornography websites have attempted to alter their percentage of pornographic content by uploading non-pornographic content to their video libraries.

⁸For example, in Texas and Louisiana, these civil penalties start at \$5,000-\$10,000 per day.

2.2 Responses to Age Verification Laws

There have been a range of responses to age verification laws from pornography sites. In my paper, I distinguish between three types of responses: websites that block access to users in states who pass age verification laws, websites that comply with age verification laws, and websites that do not comply with age verification laws. Pornhub, the most popular pornography site in the world, chose to block access to its site for users in states with age verification laws.⁹ Before blocking access to its site, Pornhub issues a warning to all its users informing them of the exact date on which they will lose access. An example of one of these warnings is shown in Figure 1. Xhamster is the only major site in my analysis that has chosen to comply with the laws in some states, rather than blocking access entirely. XNXX and Xvideos chose not to comply with the laws. These sites are not headquartered in the United States, which makes it difficult to bring a case against them in the US court system. For now, these sites have decided to remain operating as usual.

There have also been legal responses to age verification laws. The most prominent response has been from the Free Speech Coalition (FSC). The FSC is a trade organization that represents the adult industry. They have filed a number of lawsuits against state legislatures, claiming that age verification laws violate first amendment rights. The most prominent one of these cases is *Free Speech Coalition, Inc v. Paxton (Texas)*. This case made it to the Supreme Court, and in June, 2025, the Court upheld the law, affirming the government’s interest in protecting minors from accessing adult content. This court case established important precedent for the legality of age verification laws across the country.

3 Data

To estimate the impact of age verification laws on pornography consumption I use data from Google Trends. Google Trends provides a measure of a search term’s popularity within a specified time frame and region. Google Trends does not provide a measure of search *volume*. The data are normalized and presented on a scale from 0-100, where the maximum value of 100 is assigned to the data point in which the search term reaches its peak interest for the selected time frame and region. Once the peak point is identified, all other data points in this time frame and region are scaled relative to the peak. The scaling is accomplished by dividing the search frequency corresponding to the other data points by the search frequency of the peak point and multiplying by 100 (Initiative, 2025). The resulting index gives insight into how popular a term is at a given point relative to its peak across the entire time frame. For example, if a data point has an index value of 80, the search term was 80 percent as

⁹The only exceptions are Louisiana and Ohio. Before Louisiana enacted its age verification law, the state already had a system for digitally verifying identity through an app called LA Wallet. Pornhub partnered with the third party age verification platform that runs LA Wallet to allow Louisiana residents to continue accessing their site. Ohio’s law went into effect in September, 2025. Because of the way the Ohio law is written, Pornhub claims it is not affected by the law. The law does not apply to “interactive computer services,” or services in which third-party content is published to its site.

popular at that point compared to its peak. From here on out, I refer to the value of the Google Trends index as “search interest” in a term.

I collect data on search interest from all 50 states for the following terms: Pornhub, XNXX, Xvideos, Xhamster, porn, and VPN. The first four terms are the names of the top four pornography sites. Combined, these four sites receive approximately 2 billion monthly visits in the United States (Semrush, 2025). I also include search interest for the term “porn” to investigate whether there are changes in general interest for pornographic content. I include “vpn” to investigate whether there is increased interest in circumventing the law by changing the location of one’s IP address.

For my main specification, which compares early adopting states to later adopting states, I download the search interest data for each term and state separately over the period January 2, 2022 to August 2, 2024.¹⁰ This is to ensure the data for the later adopting states are not contaminated by including the period in which age verification laws are effective in these states. For alternative specifications, I extend the period to October 3, 2025. Figure 3 below shows an example of how the data are collected from Google Trends.

Although Google Trends does not provide actual search volume, it is still an insightful data source for understanding how pornography consumption responds to age verification laws. One advantage of Google Trends data in this setting is the high frequency and granularity of the data: I observe data at the weekly frequency for every state. The high frequency and granularity of Google Trends data allow me to pin down the response within a narrow window of the effective date, further supporting a causal interpretation of my results. Google Trends also has an advantage over survey data, which is typically used to measure pornography consumption in the social sciences. Survey questions that ask about pornography consumption tend to under-report the incidence and frequency, as this is often an uncomfortable topic to discuss. In contrast, Google Trends data come from a random sample of actual internet searches, making it an ideal data source for gaining insight into actual internet behavior, especially socially undesirable behavior (Stephens-Davidowitz, 2014).

4 Descriptive Evidence

In this section, I present descriptive evidence of the impact of age verification laws on pornography consumption by plotting two different time series for search interest in pornography-related terms. The first time series, displayed in Panel (a) of Figures 4-9, plots the evolution of weekly average search interest for pornography-related terms among early adopters. I plot the average search interest for the weeks leading up to and following a law’s effective

¹⁰I choose January 2, rather than the first of the year, because Google Trends applied an improvement to their data collection methodology on January 1, 2022. Search interest for all terms across all states spikes on January 1, 2022 due to this change in methodology. By excluding this one day, I ensure that the data I collect use the same methodology, and I am not picking up a spurious spike in interest due to the methodology change.

week, denoting this effective week as week zero. The second time series plots the monthly average search interest in pornography-related terms for early versus later adopters over the entire analysis period.¹¹ Taken together, the figures in this section support the narrative that, while blocked (Pornhub) and compliant (Xhamster) sites experience sharp and sustained declines in search interest following age verification laws, these declines may reflect substitution towards non-compliant sites (XNXX and Xvideos) or the use of a VPN, rather than actual declines in pornography consumption.

4.1 Search Interest in Blocked or Compliant Sites

Panel (a) of Figures 4 and 5 show relatively strong and stable search interest for blocked and compliant sites for nearly the entire pre-period. Then, in the week of enactment, search interest immediately plummets, remaining below the pre-period average for the entire post-period. These figures also highlight the advantage of the high frequency Google Trends data by showing how the striking changes in search interest align exactly with the laws’ effective dates. The drop in search interest is especially stark for Pornhub: the average value of the Google Trends index in the post-period is less than half of the average of the index in the pre-period. The series for Xhamster shows a similar, but more muted pattern. This is likely because Xhamster receives much less traffic than Pornhub, which introduces more noise in the Google Trends series. Also, Xhamster is a compliant site, so users can still access the site in states that have passed age verification laws as long as they provide the required information.¹²

The drops in search interest in Pornhub are even apparent in the raw time series shown in Panel (b) of Figure 4. In the period before any age verification law goes into effect, the series for early adopters and later adopters track each other almost perfectly. The first deviation occurs when Louisiana’s age verification law goes into effect in January, 2023. Here, the series for early adopters begins to drop away from the series for later adopters. Similar drops occur in May, 2023, July, 2023, and January, 2024. All of these dates align with the effect dates for the early adopting states. In contrast, the series for later adopters remains relatively constant over the analysis period. The immediate reactions to the laws are not as clear in Panel (b) of Figure 5 for Xhamster. However, over time, the series for the early adopters does drop away from the series for the later adopters as more laws become effective, and the later adopter series remains relatively constant.

¹¹The weekly calendar time series are available in the Appendix.

¹²Another difference between Pornhub and Xhamster is that the series for Pornhub peaks in the week just prior to the effective date, whereas the series for Xhamster peaks in the week of the effective date. This is likely because Pornhub warned users in states that passed age verification laws that they would lose access to the site when the laws were scheduled to go into effect. In the Appendix, I include an example of one of these warnings.

4.2 Search Interest in Non-Compliant Sites

As with the blocked or compliant sites, Panel (a) of Figures 6 and 7 show relatively stable search interest for the non-compliant sites for the entire pre-period. However, immediately when the laws become effective in week zero, search interest for XNXX and Xvideos spikes, remaining elevated for the entire post-period. The average value of the post-period increases by 24% compared to the pre-period average for XNXX, and by 46% for Xvideos.

Panel (b) of Figures 6 and 7 shows how the search interest in XNXX and Xvideos evolves over the entire analysis period for early adopters and later adopters. Among early adopters, search interest in XNXX and Xvideos begins its ascent after January, 2023, when the first age verification law goes into effect. Search interest in both sites continues to climb for the remainder of the analysis period. The steady climb in search interest is more apparent in Figure 7 for Xvideos, likely because Xvideos generally receives more traffic than XNXX.

One thing to note about Panel (b) in Figures 6 and 7 is that the time series for the later adopters is above the series for early adopters for the majority of the analysis period. This is an artifact of the Google Trends normalization. As shown in Panel (a) of Figures 6 and 7, for early adopters, search interest in XNXX and Xvideos spikes after the effective date. In fact, for all early adopters, search interest for XNXX and Xvideos reaches its peak after the effective date of the law. Because Google Trends normalizes all other points in the series relative to this peak, index values in the pre-period must be mechanically lower than this peak. This leads to a relative down-weighting of the pre-period index values for the early adopting states. The later adopting states do not experience the same spike in search interest, so the entire series is not reweighted as it is with early adopting states. So, while the search interest levels are mechanically different due to the Google trends normalization, the shape of the series for early adopting and later adopting states track each other nearly perfectly until the first age verification law becomes effective.

4.3 Search Interest in VPNs and Porn

Regarding VPNs, Panel (a) of Figure 8 shows relatively stable search interest in the pre-period for early adopting states. This series is a bit noisier than those in the previous sections because VPN is a lower search volume term compared to the pornography sites. However, on the week the law is enacted, there is an unmistakably large spike in search interest: for nearly every early adopter, VPN reaches its peak in search interest in the week the law goes into effect. Interest remains elevated for a few months after enactment, but quickly declines after the initial peak and seems to be approaching the pre-period average at the end of the analysis period. This is consistent with a story in which users attempt to access a blocked or compliant site, realize they are unable to or must provide personally identifying information, and then search for a way to circumvent this by using a VPN. Once a VPN is installed, users do not need to continue to search for it every time they activate it.

Panel (a) of Figure 9 shows a similar pattern for the term porn. The series is relatively flat in the pre-period, with a small but noticeable spike in search interest on the week the law goes into effect. However, this elevated interest is short-lived and the series returns to its pre-period levels after about the first month following the effective date. This is consistent with a story in which users attempt to visit a blocked or compliant site, but then decide to find an alternative site by performing a general google search for “porn.” This story could also be consistent with people searching for information about the law after it becomes effective, as these searches would likely contain the word “porn” in them.

Because the peaks for VPN and porn are highly localized and short-lived, Panel (b) of Figures 8 and 9 do not depict major differences in search interest between early and later adopters. In levels, the later adopters appear to have slightly higher search interest in VPNs than early adopters, especially in the beginning of the analysis period. This is likely due to the mechanical down-weighting of the pre-period for early adopters. VPN is a relatively low-volume search term, so the massive spike in interest after the laws’ effective dates causes significant compression of the index values in other points of the analysis period. Overall, the time series for early and later adopters for both terms is relatively flat over the entire analysis period.

5 Empirical Strategy

To estimate the impact of age verification laws on pornography consumption, I use a difference-in-differences framework that compares outcomes in early adopting states relative to later adopting states before and after the effective dates of the age verification laws.¹³ I rely on an event study framework for my primary specification to capture the dynamic effects of treatment. I estimate the event study for each search term separately. The event study specification is:

$$y_{st} = \alpha_s + \gamma_t + \sum_{j=-12}^{j=-2} \beta_j \times 1(\text{Early})_s \times D_{s,t-j} + \sum_{j=0}^{j=6} \beta_j \times 1(\text{Early})_s \times D_{s,t-j} + \varepsilon_{st},$$

where y_{st} represents the search interest for a term in state s in time t . The specification includes state (α_s) and calendar time (γ_t) fixed effects. Calendar time is measured at the month-year level.¹⁴ $1(\text{Early})_s$ is an indicator equal to one if a state is an early adopter.

¹³The early adopters in my sample are Louisiana, Utah, Mississippi, Virginia, Arkansas, Montana, and North Carolina. The late adopters are Indiana, Alabama, Oklahoma, Florida, South Carolina, Tennessee, Georgia, Wyoming, South Dakota, North Dakota, Arizona, and Missouri. I exclude Texas from the early adopter sample because the legal complications make it unclear when the law really took effect. I also exclude Idaho, Kansas, Kentucky, and Nebraska from the late adopters sample to avoid contamination of the control group, as these states pass laws within the six month post period.

¹⁴Though I observe Google Trends data at the weekly frequency, I aggregate to the monthly level for

$D_{s,t-j}$ is an indicator for event time, j , meaning that the effective date for a law in state s is j periods before this observation’s calendar time, t . The late adopting states have $D_{s,t-j} = 0$ for every period, t , as they do not enact laws within the time frame considered for estimation. I split the summations into two terms representing the lags and leads of the event time, respectively. This split also makes explicit that the reference period is period $j = -1$, the month before the law takes effect. The coefficients of interest are the β_j s for $j \geq 0$, which capture the dynamic effects of treatment: the differential change in a term’s search interest for early adopting versus late adopting states at a given period, j , relative to the omitted period, β_{-1} . I cluster the standard errors at the state level, as this is the level of treatment. I discuss the results from estimating this event study Section 6. Figures 10-14 display the estimates of each β_j along with 95% confidence intervals.

Given the rich variation in event time and treatment status (illustrated in Table 1), there are several other specifications that could be appropriate for my setting. In Section 9.5.1 I present the results for an alternative specification in which I use only ever-treated states, leveraging variation in event time for identification. In Section 9.5.2 I present the results for an alternative specification in which I use a difference-in-differences framework comparing states who ever adopt age verification laws to states who never adopt age verification laws.

I choose the framework that compares early adopters to later adopters for my main specification because of the advantages it has over the other two approaches. The early versus later adopters approach has an advantage over the timing based approach because it allows for estimation based on a balanced panel. In the timing-based approach, the control group evolves over the analysis period as more states adopt age verification laws, and each treated state is treated for varying durations. The early versus late adopters approach has advantage over the adopters versus never adopters approach because later adopters are likely a more valid counterfactual than states who never adopt age verification laws.

6 Results

This section largely formalizes the insights from the descriptive evidence in Section 4 above by presenting the differences between early and later adopting states in an event study framework. I confirm that the observed divergence in search interest between early adopters and later adopters is statistically significant for the entire post period for Pornhub, XNXX, and Xvideos. I also confirm that the short-term spikes in search interest for VPN and porn are statistically significant. The results for Xhamster are descriptively similar to the observations from Section 4, but they are insignificant because the lower volume of this term introduces noise in estimation. Figures 10-14 display the results discussed in this section. In each figure, the point estimates for each β_j are reported alongside the 95% confidence intervals. The omitted period is the month before the law went into effect.

greater precision in my primary specification. Event studies at the weekly frequency can be found in the appendix.

One thing to note is that the Google Trends normalization introduces a few quirks regarding the pre-period coefficients in the event studies. As shown in Panel (b) of Figures 6 and 7, for terms such as XNXX and Xvideos that become relatively more popular after the effective date and hit their peak in the post-period, the pre-period index is down-weighted relative to this post-period peak among early adopting states. Later adopting states that do not have sharp changes in search interest over the relevant time period do not experience this down-weighting of the index values. The problem of down-weighting manifests as negative, statistically significant coefficients in the pre-period of the event studies for XNXX and Xvideos, as shown in Figures 12 and 13, respectively. The primary difference driving this spuriously negative effect is the normalization-induced difference in pre-period levels between early adopters and later adopters.¹⁵

A more complex quirk occurs because of the interaction between the Google Trends normalization and the addition of state fixed effects. Google Trends already normalizes each state’s series to its within-state maximum value, and state fixed effects demean the search interest in each term by its within-state average over the analysis period. A term such as Pornhub that experiences a sharp drop in search interest after the law goes into effect significantly brings down the within-state average for early adopting states. So, when each term is demeaned by its within-state average, these lower means raise the pre-period residual for early adopting states, making the pre-period appear elevated relative to later adopting states, even when the actual pre-trend difference is negligible, as shown in Panel (b) of Figure 4. This problem of raising the pre-period residual for early adopting states manifests as positive pre-period coefficients in the event study for Pornhub, as shown in Figure 10. This difference is not driven by differences in levels between early adopters and later adopters in the pre-period: it is driven by the within-state demeaning.¹⁶

In summary, the significance of these pre-period coefficients for Pornhub, XNXX, and Xvideos, is an artifact of the normalization of Google Trends data and does not reflect actual behavioral differences between the early adopters and later adopters in the pre-period. In the following subsections, I describe the results in more detail for each term in my analysis.

¹⁵One alternative I explored to mitigate the problem of down-weighting the pre-period index values is to download the search interest data for each term along with another neutral high volume term (such as “time” or “weather”). This effectively indexes the search term of interest to the peak point for the neutral term. There are still many problems with this approach. The first is that it is difficult to find a neutral high volume term that, for all states and search terms, is 1) sufficiently high volume to serve as the reference term, and 2) is not so high volume that the time series for the term of interest collapses to a flat line when the high volume term is added for comparison. More importantly, the peak time period for these neutral terms often varies by state. Even for terms such as “time” that tend to have the same peak across states (due to time changes in the spring and fall), the approach still suffers from the problem of some terms reaching the peak in a state’s pre-period and others in the state’s post-period. Thus, this approach does not solve the issue and leads to a non-intuitive interpretation of the index that varies from state to state.

¹⁶In fact, the size of the pre-period coefficients for Pornhub is almost exactly equal to the difference in the average search interest across the entire analysis period for early versus later adopting states. Moreover, when I exclude state fixed effects from the event study specification for Pornhub, the pre-period coefficients are virtually zero and precisely estimated. In the Appendix, I include, for each term, a side by side comparison of the results from the main specification with both state and calendar time fixed effects and a specification where I exclude state fixed effects.

6.1 Effects on Search Interest in Blocked or Compliant Sites

Figure 10 reports the event study results for the effects of age verification laws on search interest in Pornhub. This figure shows that the difference in search interest between early and later adopters is flat and stable for the entire pre-period. However, once the law goes into effect in period zero, there is a sharp drop in search interest for early adopters. This drop in search interest is statistically significant as early as one month after the law goes into effect.¹⁷ The post-period estimates indicate that the average search interest in Pornhub for early adopting states is more than 20 points lower than the average search interest in later adopting states relative to the omitted period. This difference in search interest between early and later adopters is sustained throughout the entire post-period.

Figure 11 reports the event study results for the effects of age verification laws on search interest in Xhamster. This figure shows that differences in pre-period search interest between early and later adopters is relatively flat and insignificant. When the law goes into effect in period zero, there is visual evidence of a small, statistically insignificant spike in search interest among early adopters, mirroring the descriptive evidence from Panel a) of Figure 5. There is also visual evidence of a decline in search interest among early adopters after this peak in period zero, but the post-period coefficients only become marginally significant in the last few months of the analysis period. As mentioned previously, the lack of significance is likely due to the fact that search interest in Xhamster is more noisily measured than the other three pornography sites.

6.2 Effects on Search Interest in Non-Compliant Sites

Figures 12 and 13 report the event study results for the effects of age verification laws on search interest in XNXX and Xvideos, respectively. Both figures show flat, stable differences in search interest between early and later adopters. However, once the law goes into effect in period zero, there is an increase in search interest for both XNXX and Xvideos among early adopters. This increase is statistically significant as early as one month after the law goes into effect.¹⁸ The post-period estimates indicate that average search interest in XNXX for early adopting states is more than 10 points higher than the average search interest in later adopting states relative to the omitted period. The difference for Xvideos is more than 15 points on average. This difference in search interest between early and later adopters is sustained throughout the entire post-period.

¹⁷In fact, the weekly event study in Figure 22 shows that this drop is significant after the second week the law goes into effect.

¹⁸The weekly event studies in Figures 25 and 24 show that this increase is statistically significant as early as the second week after the laws go into effect.

6.3 Effects on Search Interest in VPNs and Porn

Figures 14 and 15 report the event study results for the effects of age verification laws on search interest in VPNs and porn, respectively. These figures show that the difference in search interest between the early and later adopters is relatively flat and insignificant in the pre-period, though the estimates for VPN are a bit noisier due to the lower search volume of this term. Then, in period zero, both terms experience an immediate spike in search interest among early adopters that is statistically significant even in month zero.¹⁹ This peaked search interest declines quickly for both, but remains significant for VPN four months after the law goes into effect.

7 Discussion

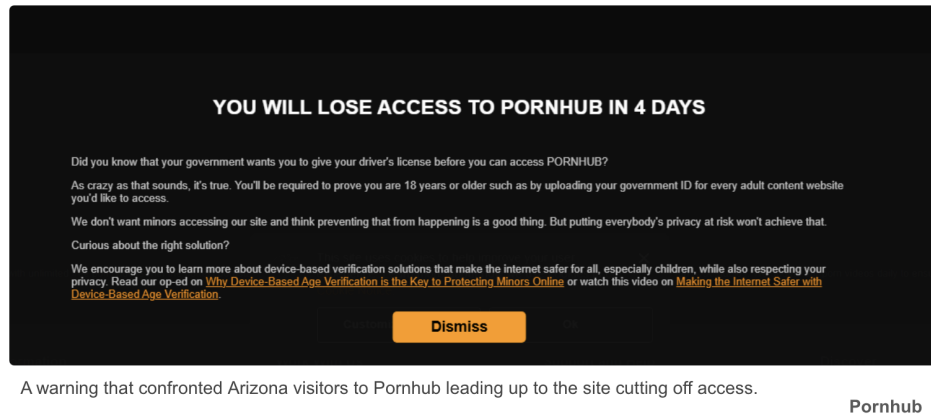
Using Google Trends data, I present evidence that, following the enactment of age verification laws, there is increased interest in substituting to non-compliant sites. There is also evidence of potential circumvention of the law by using a VPN. Because Google Trends data is a normalized index, there are some limitations in using these results to make statements about the magnitudes of the effects. Google Trends data is also only a measure of search behavior, and it may miss important behavioral responses for those who are accessing pornography sites directly (rather than through a search engine). Because of these limitations, I am currently working on obtaining access to additional data that will allow me improve on the Google Trends analysis by looking at website traffic and other direct measures of internet behavior.

¹⁹The weekly event studies in Figures 26 and 27 show that these spikes are significant in week zero, lining up exactly with the period in which the laws go into effect.

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8 Tables and Figures



A warning that confronted Arizona visitors to Pornhub leading up to the site cutting off access.

Pornhub

Figure 1: Warning Issued by Pornhub before Age Verification Law Takes Effect

Notes: This screenshot shows an example of the warnings Pornhub issues to users in states that have passed age verification laws. This example is from Arizona.

Source: L'Heureux, T.J. "Pornhub Has Cut Off Arizona — There's a Way Around It." Phoenix New Times, November 3, 2025. <https://www.phoenixnewtimes.com/news/how-to-use-a-vpn-to-access-pornhub-in-arizona-22736631/>

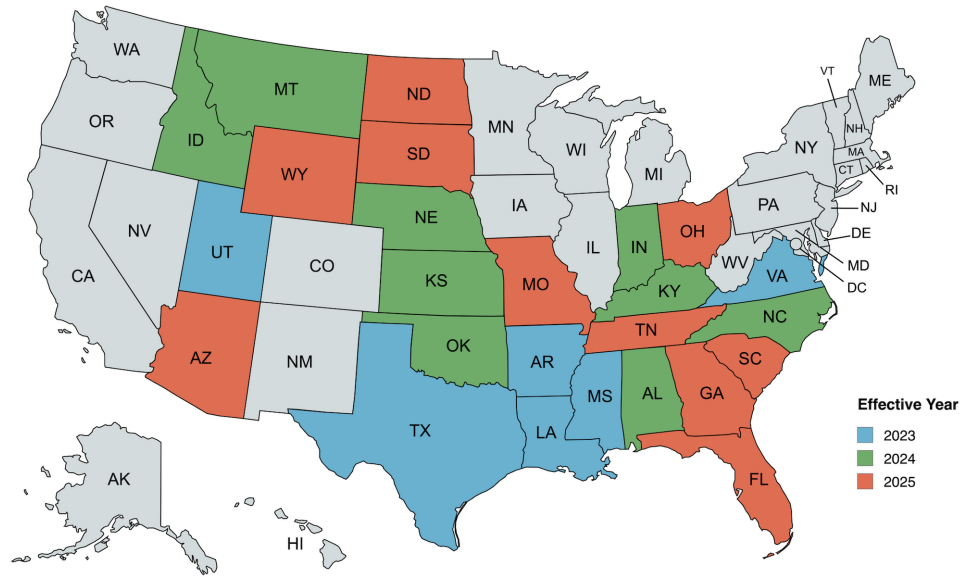


Figure 2: States with Age Verification Laws

Notes: This figure shows which states have passed age verification laws and the year the laws went into effect. States shaded in blue had age verification laws with effective dates in 2023 (Louisiana, Utah, Mississippi, Virginia, Arkansas, and Texas). States shaded in green had effective dates in 2024 (Montana, North Carolina, Idaho, Kansas, Kentucky, Nebraska, Indiana, Alabama, and Oklahoma). States shaded in orange had effective dates in 2025 (Florida, South Carolina, Tennessee, Georgia, Wyoming, South Dakota, North Dakota, Arizona, Ohion, and Missouri– scheduled to take effect at the end of November).

Table 1: List of State Age Verification Laws with Passage and Effective Dates

State (1)	Title (2)	Passed On (3)	Effective On (4)
Louisiana	HB 142 (2022)/ Act 440	2022-06-15	2023-01-01
Utah	SB 287	2023-03-14	2023-05-03
Mississippi	SB 2346	2023-04-18	2023-07-01
Virginia	SB 1515	2023-05-12	2023-07-01
Arkansas	SB 66	2023-04-11	2023-07-31
Texas*	HB 1181	2023-06-12	2023-09-19
Montana	SB 544	2023-05-22	2024-01-01
North Carolina	HB 8	2023-09-29	2024-01-01
Idaho	H 498	2024-03-20	2024-07-01
Kansas	SB 394	2024-04-25	2024-07-01
Kentucky	HB 278	2024-04-05	2024-07-15
Nebraska	LB 1092	2024-04-16	2024-07-19
Indiana	SB 17	2024-03-13	2024-08-16
Alabama	HB 164	2024-04-18	2024-10-01
Oklahoma	SB 1959	2024-04-26	2024-11-01
Florida	HB 3	2024-03-25	2025-01-01
South Carolina	HB 3424	2024-05-21	2025-01-01
Tennessee	SB 1792	2024-05-28	2025-01-01
Georgia	SB 351	2024-04-23	2025-07-01
Wyoming	HB 43	2025-03-13	2025-07-01
South Dakota	HB 1053	2025-02-27	2025-07-01
North Dakota	HB 1561	2025-04-14	2025-08-01
Arizona	HB 2112	2025-05-13	2025-09-25
Ohio	HB 96	2025-06-30	2025-09-30
Missouri	15 CSR 60-18	2025-05-15	2025-11-30

Notes: This table lists all states who have passed age verification laws as of October, 2025. Column 2 lists the title of the law when it was passed. Column 3 lists the date the law was passed in the state legislature, while column 4 lists the date the law was scheduled to take effect. For all states but Texas, the laws were effective on the days reported in column 4.

*The effective date for Texas reflects the date originally approved by the legislature when the law was passed on June 12, 2023. On August 4, 2023, the Free Speech Coalition filed a lawsuit challenging HB 1181. Just before the law was scheduled to take effect on September 1, 2023, a federal judge in Austin, Texas issued a temporary ban on enforcing the law. In November of 2023, the temporary ban was lifted, allowing Texas to enforce the law. On February 26, 2024, Texas Attorney General, Ken Paxton, filed a lawsuit against Aylo Global Entertainment, the pornography distributing company for Pornhub, for violating HB 1181. On March 14, 2024, Pornhub blocked access to its site and affiliated sites entirely in Texas.

I source the information for this table from the Age Verification Providers Association (AVPA). More information can be found at <https://avpassociation.com/4271-2/>.

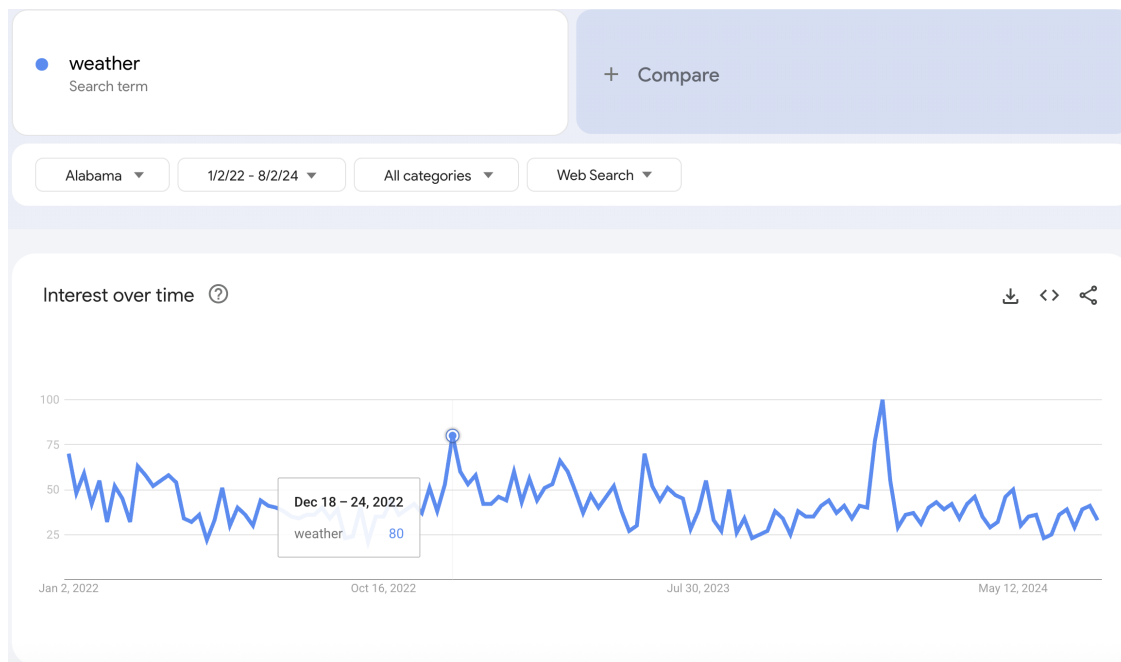
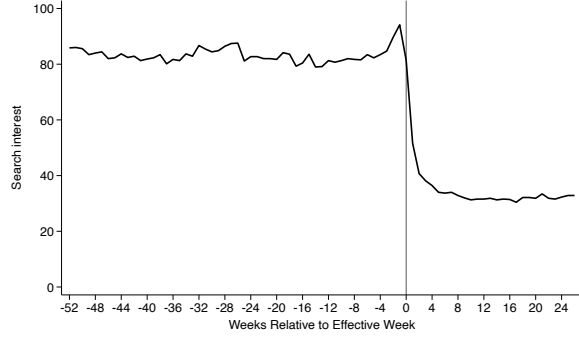
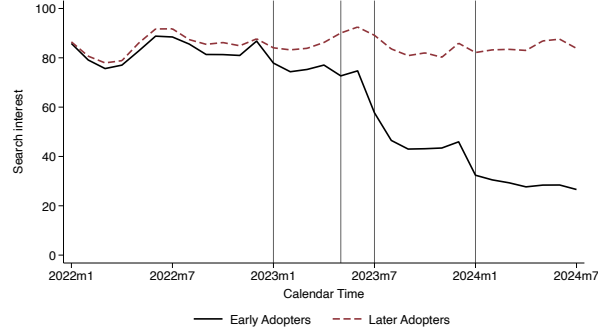


Figure 3: Example of Google Trends Data Collection

Notes: This figure illustrates how I collect the data from Google Trends. In the top left corner, I input the term of interest (“weather” in this example). Just below the box for the search term, I select the region of interest (“Alabama” in this example). Then, I select the time frame. For my primary specification, the time frame is January 02, 2022 to August 2, 2024. I download the data for every search term and every state in my analysis sample, keeping the time frame constant. In the example above, the search term “weather” reaches its peak in January, 2024. The data point singled out in December, 2022 has an index value of 80, meaning that searches for “weather” were 80 percent as popular in December, 2022 as they were on the peak day in January, 2024.



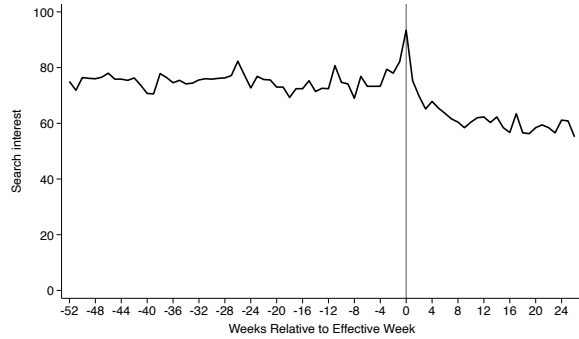
(a) Relative Week Time Series for Early Adopters



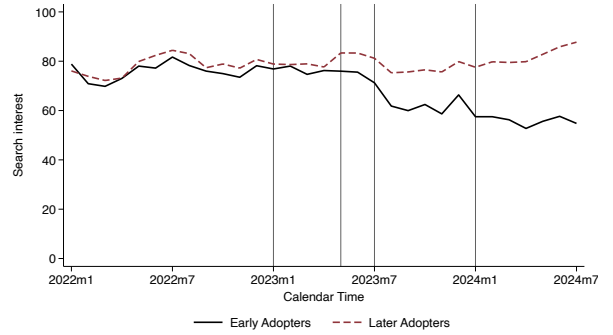
(b) Calendar-Month Time Series for Early versus Late Adopters

Figure 4: Search Interest in Pornhub

Notes: This figure shows the raw means for search interest in Pornhub among the early adopters and later adopters in the sample period: January 2, 2022 to August 2, 2024. Panel (a) reports the weekly average search interest in Pornhub for all early adopting states in my sample. The x-axis for Panel (a) is the week relative to the effective date of the law. I highlight the effective date of the law with a vertical line at week zero. Panel (b) reports the monthly average search interest in Pornhub for early adopters (solid black line) and later adopters (dashed red line). The x-axis for Panel (b) is calendar time in months. Each vertical line in Panel (b) corresponds to an effective date for the early adopting states: Louisiana in January, 2023; Utah in May, 2023; Mississippi, Virginia, and Arkansas in July 2023, and Montana and North Carolina in January, 2024.



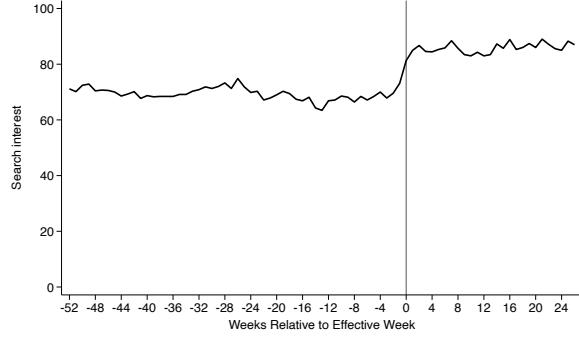
(a) Relative Week Time Series for Early Adopters



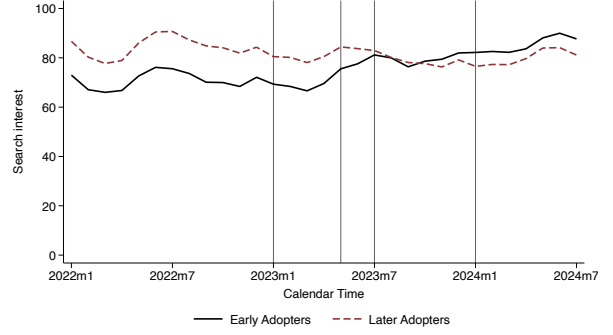
(b) Calendar-Month Time Series for Early versus Late Adopters

Figure 5: Search Interest in Xhamster

Notes: This figure shows the raw means for search interest in Xhamster among the early adopters and later adopters in the sample period: January 2, 2022 to August 2, 2024. Panel (a) reports the weekly average search interest in Xhamster for all early adopting states in my sample. The x-axis for Panel (a) is the week relative to the effective date of the law. I highlight the effective date of the law with a vertical line at week zero. Panel (b) reports the monthly average search interest in Xhamster for early adopters (solid black line) and later adopters (dashed red line). The x-axis for Panel (b) is calendar time in months. Each vertical line in Panel (b) corresponds to an effective date for the early adopting states: Louisiana in January, 2023; Utah in May, 2023; Mississippi, Virginia, and Arkansas in July 2023, and Montana and North Carolina in January, 2024.



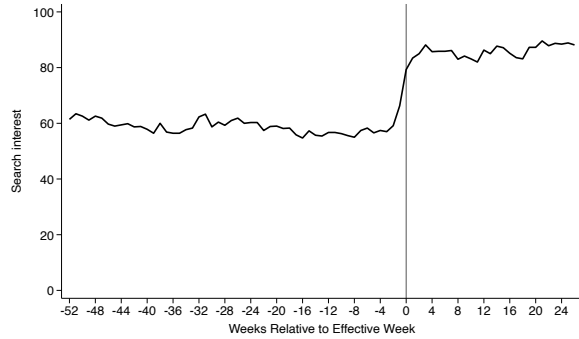
(a) Relative Week Time Series for Early Adopters



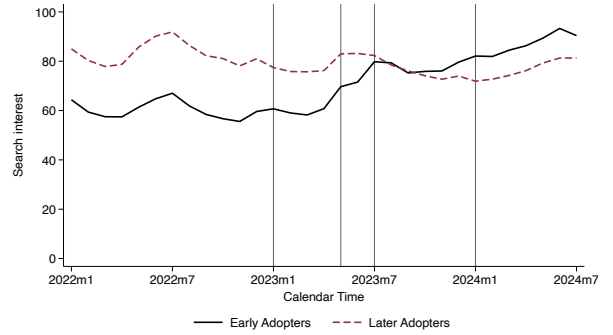
(b) Calendar-Month Time Series for Early versus Late Adopters

Figure 6: Search Interest in XNXX

Notes: This figure shows the raw means for search interest in XNXX among the early adopters and later adopters in the sample period: January 2, 2022 to August 2, 2024. Panel (a) reports the weekly average search interest in XNXX for all early adopting states in my sample. The x-axis for Panel (a) is the week relative to the effective date of the law. I highlight the effective date of the law with a vertical line at week zero. Panel (b) reports the monthly average search interest in XNXX for early adopters (solid black line) and later adopters (dashed red line). The x-axis for Panel (b) is calendar time in months. Each vertical line in Panel (b) corresponds to an effective date for the early adopting states: Louisiana in January, 2023; Utah in May, 2023; Mississippi, Virginia, and Arkansas in July 2023, and Montana and North Carolina in January, 2024.



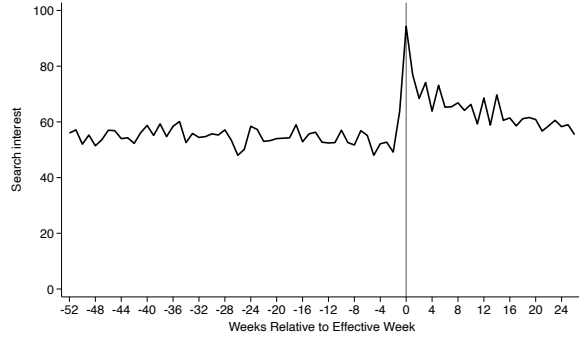
(a) Relative Week Time Series for Early Adopters



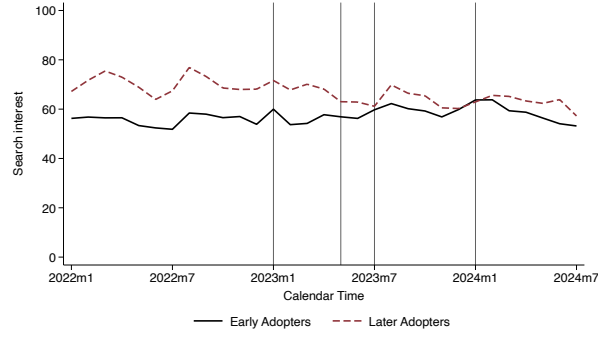
(b) Calendar-Month Time Series for Early versus Late Adopters

Figure 7: Search Interest in Xvideos

Notes: This figure shows the raw means for search interest in Xvideos among the early adopters and later adopters in the sample period: January 2, 2022 to August 2, 2024. Panel (a) reports the weekly average search interest in Xvideos for all early adopting states in my sample. The x-axis for Panel (a) is the week relative to the effective date of the law. I highlight the effective date of the law with a vertical line at week zero. Panel (b) reports the monthly average search interest in Xvideos for early adopters (solid black line) and later adopters (dashed red line). The x-axis for Panel (b) is calendar time in months. Each vertical line in Panel (b) corresponds to an effective date for the early adopting states: Louisiana in January, 2023; Utah in May, 2023; Mississippi, Virginia, and Arkansas in July 2023, and Montana and North Carolina in January, 2024.



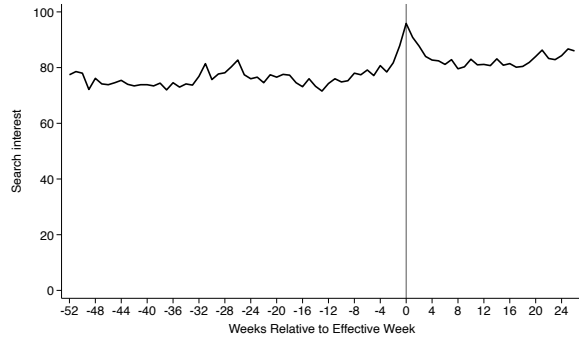
(a) Relative Week Time Series for Early Adopters



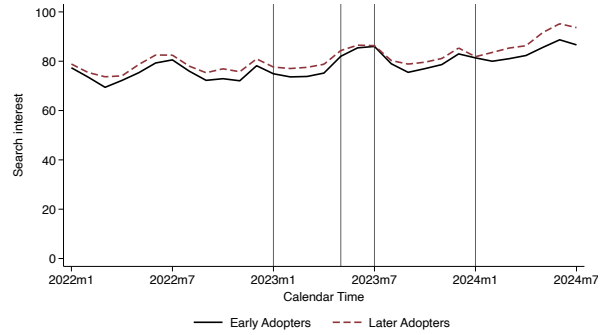
(b) Calendar-Month Time Series for Early versus Late Adopters

Figure 8: Search Interest in VPN

Notes: This figure shows the raw means for search interest in VPN among the early adopters and later adopters in the sample period: January 2, 2022 to August 2, 2024. Panel (a) reports the weekly average search interest in VPN for all early adopting states in my sample. The x-axis for Panel (a) is the week relative to the effective date of the law. I highlight the effective date of the law with a vertical line at week zero. Panel (b) reports the monthly average search interest in VPN for early adopters (solid black line) and later adopters (dashed red line). The x-axis for Panel (b) is calendar time in months. Each vertical line in Panel (b) corresponds to an effective date for the early adopting states: Louisiana in January, 2023; Utah in May, 2023; Mississippi, Virginia, and Arkansas in July 2023, and Montana and North Carolina in January, 2024.



(a) Relative Week Time Series for Early Adopters



(b) Calendar-Month Time Series for Early versus Late Adopters

Figure 9: Search Interest in Porn

Notes: This figure shows the raw means for search interest in porn among the early adopters and later adopters in the sample period: January 2, 2022 to August 2, 2024. Panel (a) reports the weekly average search interest in porn for all early adopting states in my sample. The x-axis for Panel (a) is the week relative to the effective date of the law. I highlight the effective date of the law with a vertical line at week zero. Panel (b) reports the monthly average search interest in porn for early adopters (solid black line) and later adopters (dashed red line). The x-axis for Panel (b) is calendar time in months. Each vertical line in Panel (b) corresponds to an effective date for the early adopting states: Louisiana in January, 2023; Utah in May, 2023; Mississippi, Virginia, and Arkansas in July 2023, and Montana and North Carolina in January, 2024.

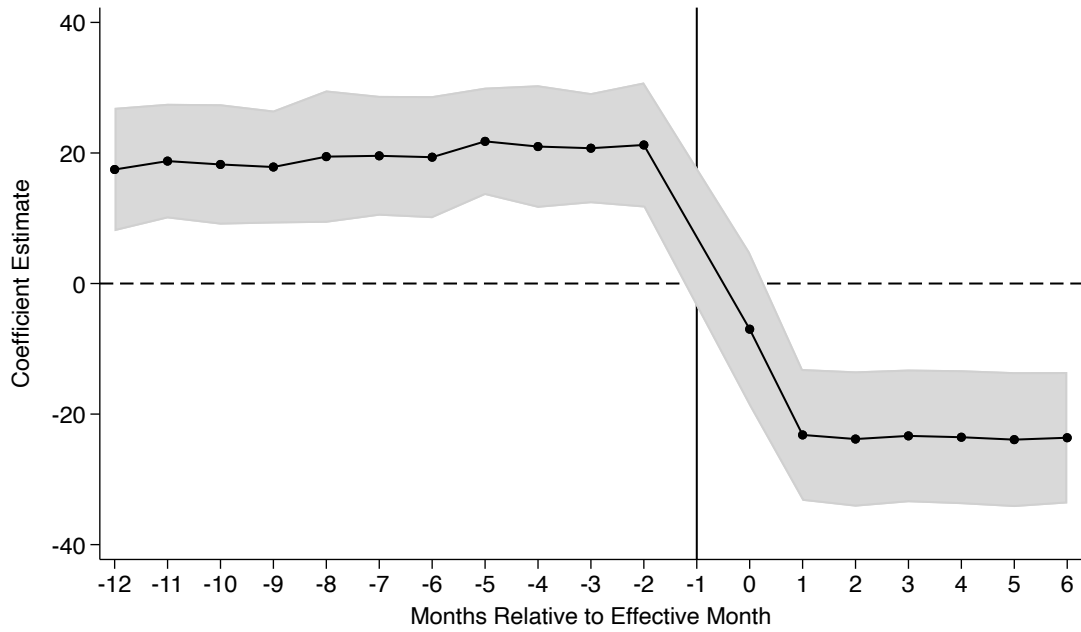


Figure 10: Effects on Search Interest in Pornhub

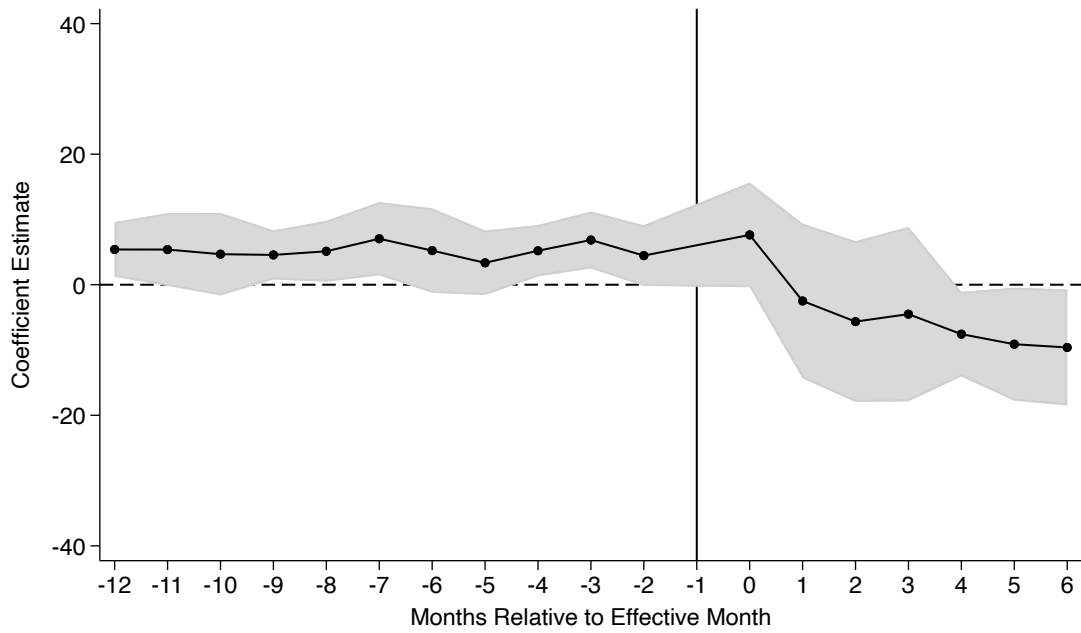


Figure 11: Effects on Search Interest in Xhamster

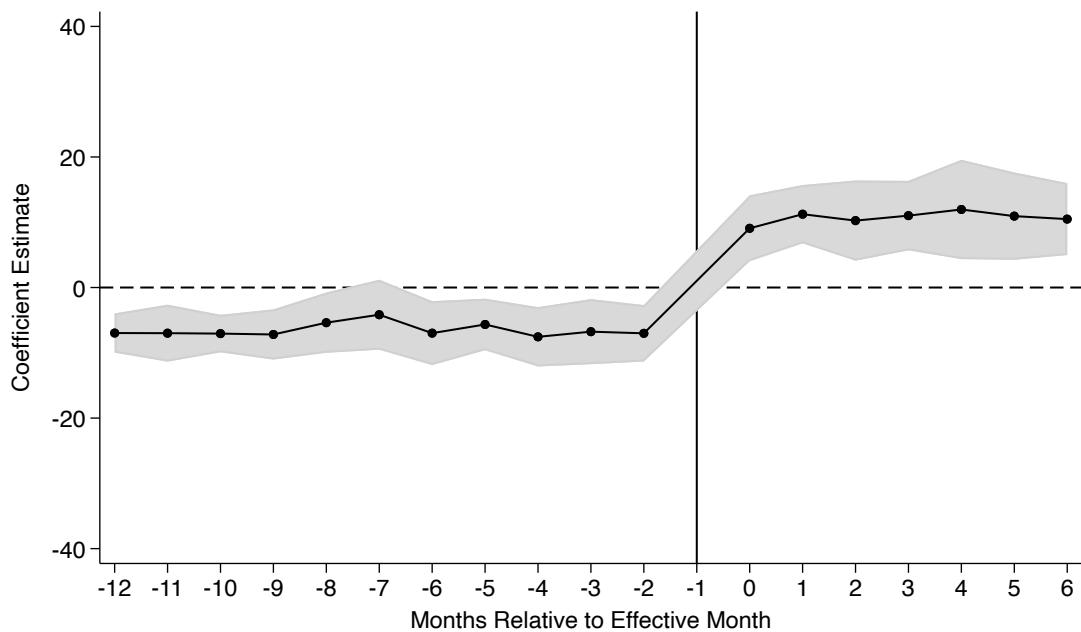


Figure 12: Effects on Search Interest in XNXX

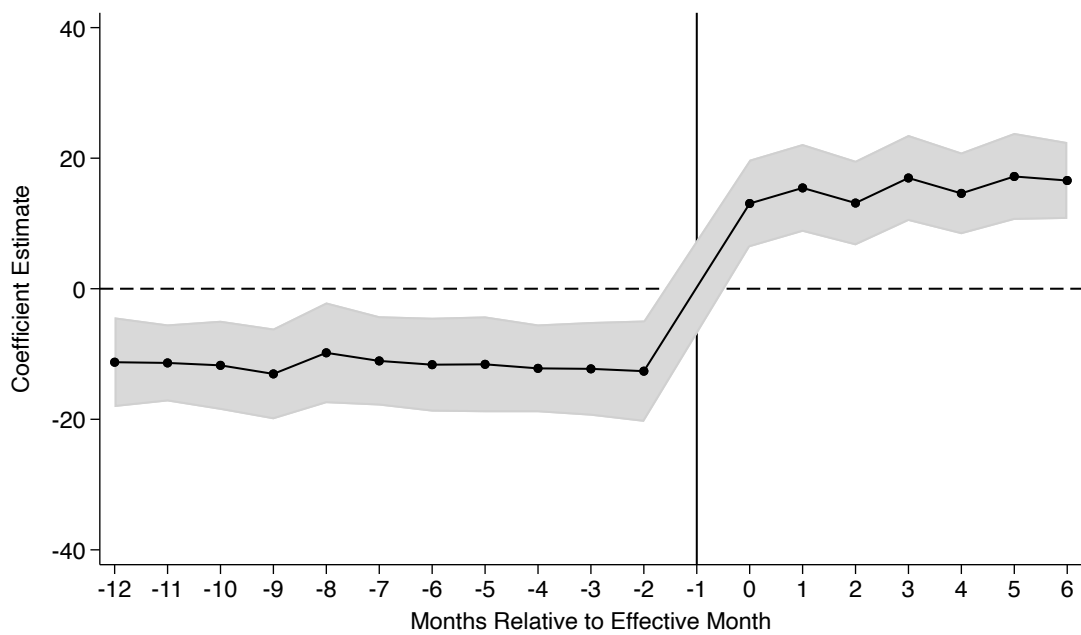


Figure 13: Effects on Search Interest in Xvideos

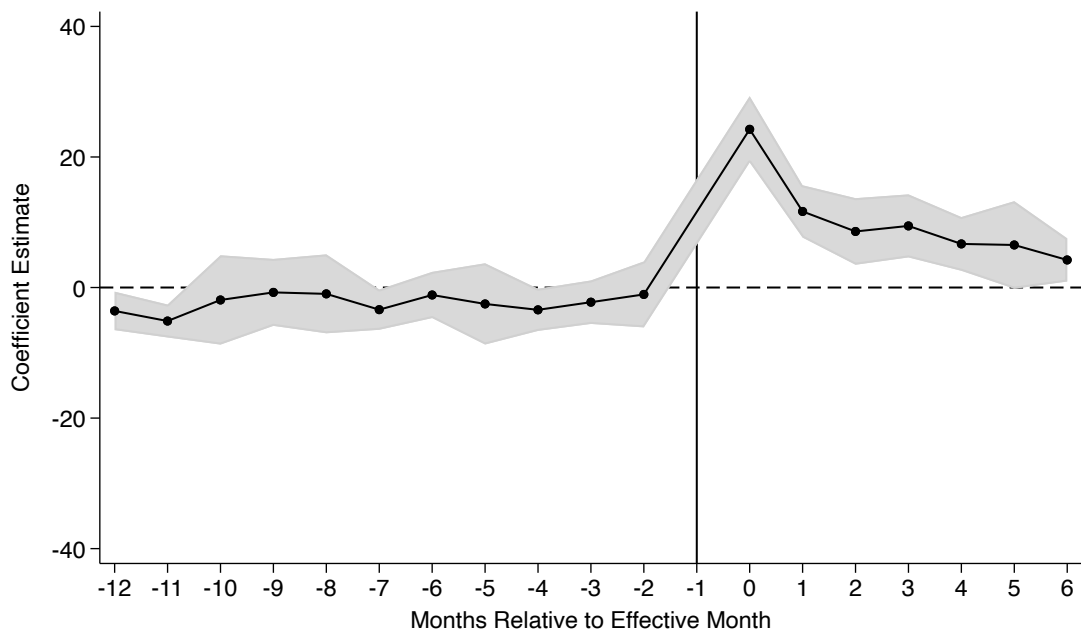


Figure 14: Effects on Search Interest in VPN

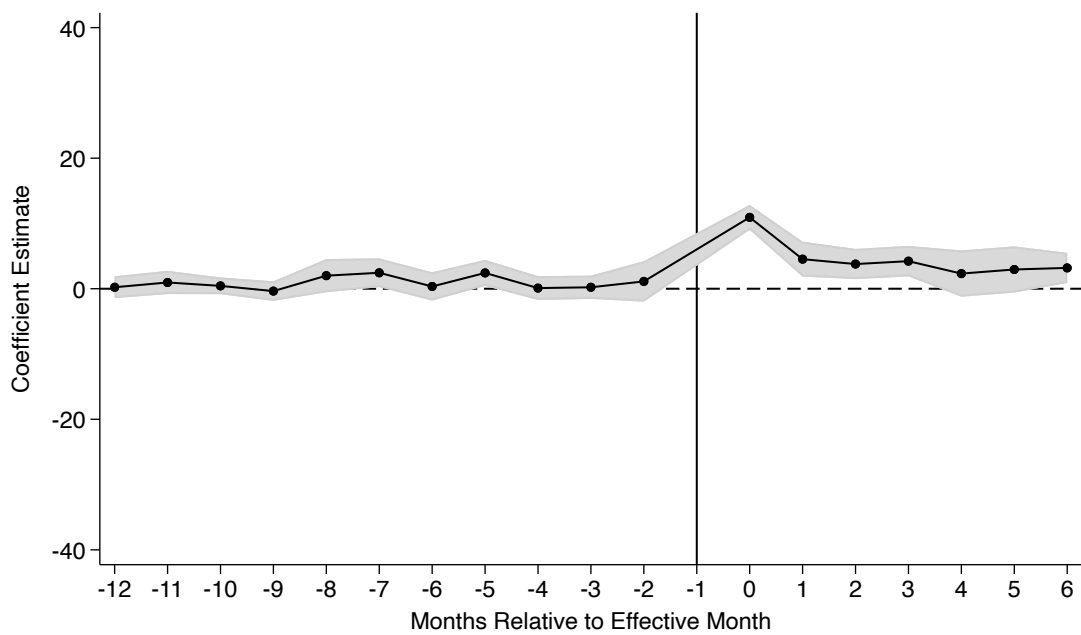
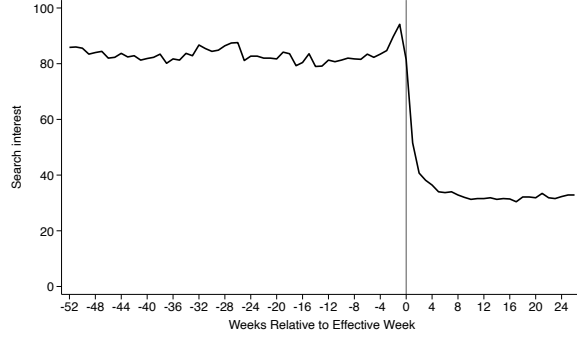


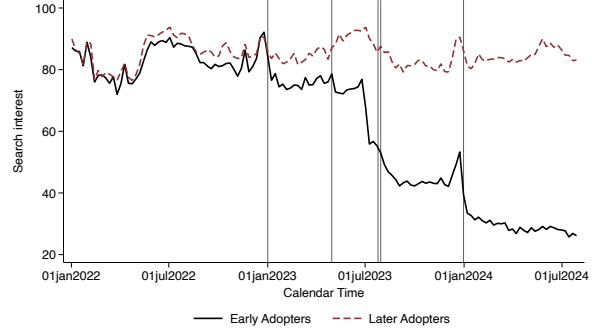
Figure 15: Effects on Search Interest in Porn

9 Appendix

9.1 Weekly Calendar Time Series



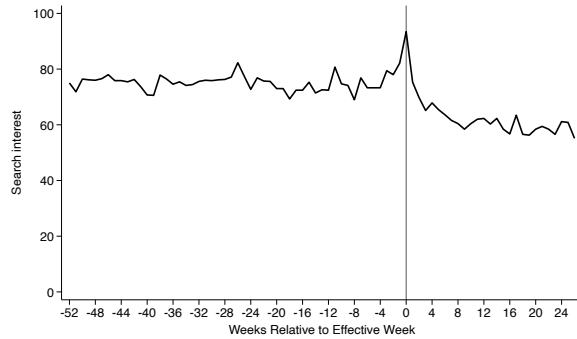
(a) Relative Week Time Series for Early Adopters



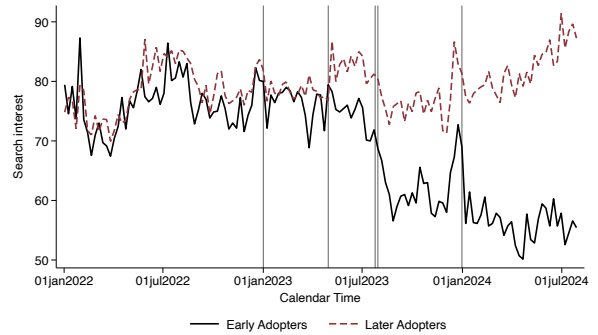
(b) Calendar-Week Time Series for Early versus Late Adopters

Figure 16: Search Interest in Pornhub

Notes: This figure shows the raw means for search interest in Pornhub among the early adopters and later adopters in the sample period: January 2, 2022 to August 2, 2024. Panel (a) reports the weekly average search interest in Pornhub for all early adopting states in my sample. The x-axis for Panel (a) is the week relative to the effective date of the law. I highlight the effective date of the law with a vertical line at week zero. Panel (b) reports the weekly average search interest in Pornhub for early adopters (solid black line) and later adopters (dashed red line). The x-axis for Panel (b) is calendar time in weeks. Each vertical line in Panel (b) corresponds to an effective date for the early adopting states: Louisiana in January, 2023; Utah in May, 2023; Mississippi, Virginia, and Arkansas in July 2023, and Montana and North Carolina in January, 2024.



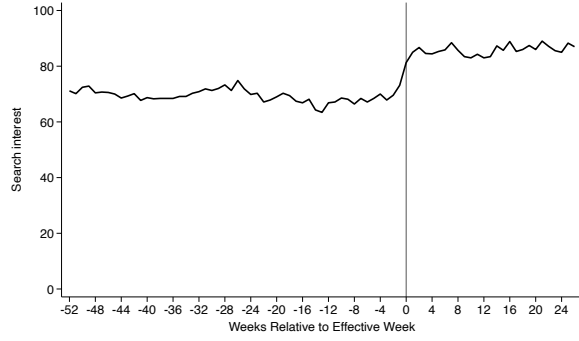
(a) Relative Week Time Series for Early Adopters



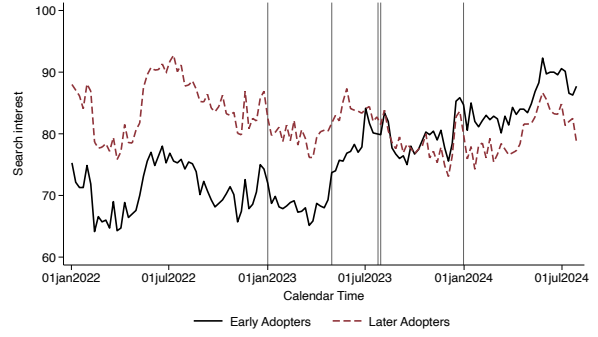
(b) Calendar-Week Time Series for Early versus Late Adopters

Figure 17: Search Interest in Xhamster

Notes: This figure shows the raw means for search interest in Xhamster among the early adopters and later adopters in the sample period: January 2, 2022 to August 2, 2024. Panel (a) reports the weekly average search interest in Xhamster for all early adopting states in my sample. The x-axis for Panel (a) is the week relative to the effective date of the law. I highlight the effective date of the law with a vertical line at week zero. Panel (b) reports the weekly average search interest in Xhamster for early adopters (solid black line) and later adopters (dashed red line). The x-axis for Panel (b) is calendar time in weeks. Each vertical line in Panel (b) corresponds to an effective date for the early adopting states: Louisiana in January, 2023; Utah in May, 2023; Mississippi, Virginia, and Arkansas in July 2023, and Montana and North Carolina in January, 2024.



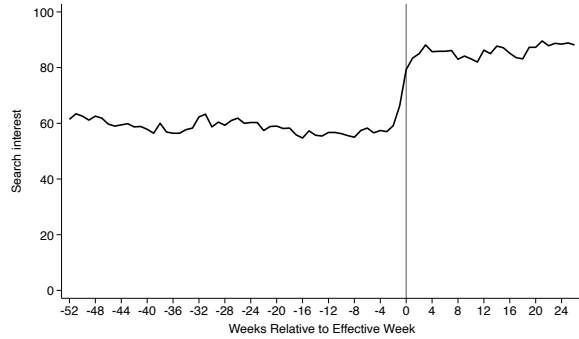
(a) Relative Week Time Series for Early Adopters



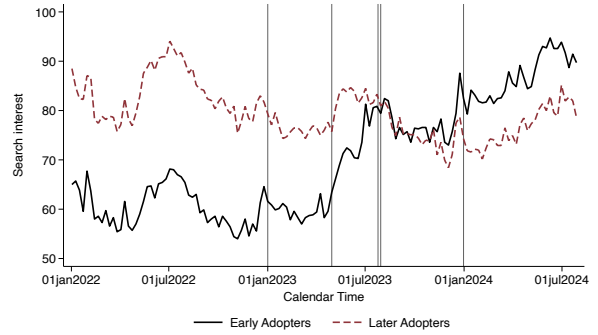
(b) Calendar-Week Time Series for Early versus Late Adopters

Figure 18: Search Interest in XNXX

Notes: This figure shows the raw means for search interest in XNXX among the early adopters and later adopters in the sample period: January 2, 2022 to August 2, 2024. Panel (a) reports the weekly average search interest in XNXX for all early adopting states in my sample. The x-axis for Panel (a) is the week relative to the effective date of the law. I highlight the effective date of the law with a vertical line at week zero. Panel (b) reports the weekly average search interest in XNXX for early adopters (solid black line) and later adopters (dashed red line). The x-axis for Panel (b) is calendar time in weeks. Each vertical line in Panel (b) corresponds to an effective date for the early adopting states: Louisiana in January, 2023; Utah in May, 2023; Mississippi, Virginia, and Arkansas in July 2023, and Montana and North Carolina in January, 2024.



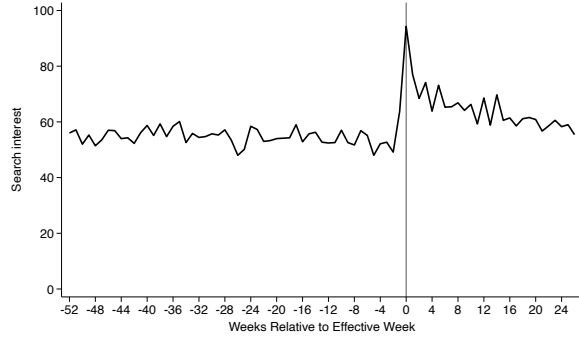
(a) Relative Week Time Series for Early Adopters



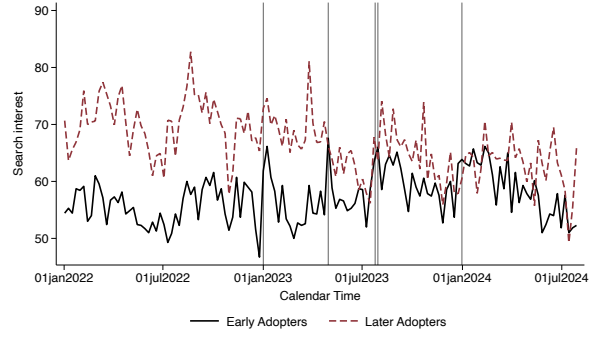
(b) Calendar-Week Time Series for Early versus Late Adopters

Figure 19: Search Interest in Xvideos

Notes: This figure shows the raw means for search interest in Xvideos among the early adopters and later adopters in the sample period: January 2, 2022 to August 2, 2024. Panel (a) reports the weekly average search interest in Xvideos for all early adopting states in my sample. The x-axis for Panel (a) is the week relative to the effective date of the law. I highlight the effective date of the law with a vertical line at week zero. Panel (b) reports the weekly average search interest in Xvideos for early adopters (solid black line) and later adopters (dashed red line). The x-axis for Panel (b) is calendar time in weeks. Each vertical line in Panel (b) corresponds to an effective date for the early adopting states: Louisiana in January, 2023; Utah in May, 2023; Mississippi, Virginia, and Arkansas in July 2023, and Montana and North Carolina in January, 2024.



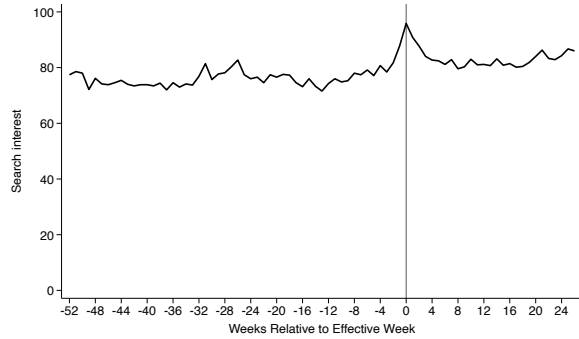
(a) Relative Week Time Series for Early Adopters



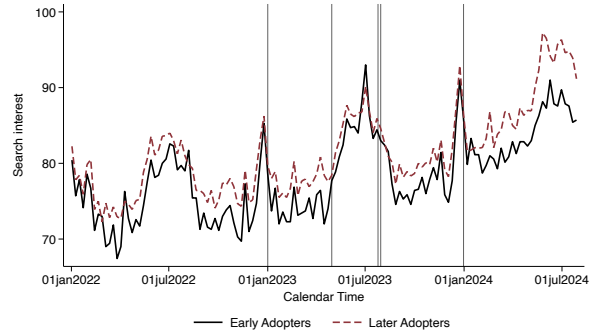
(b) Calendar-Week Time Series for Early versus Late Adopters

Figure 20: Search Interest in VPN

Notes: This figure shows the raw means for search interest in VPN among the early adopters and later adopters in the sample period: January 2, 2022 to August 2, 2024. Panel (a) reports the weekly average search interest in VPN for all early adopting states in my sample. The x-axis for Panel (a) is the week relative to the effective date of the law. I highlight the effective date of the law with a vertical line at week zero. Panel (b) reports the weekly average search interest in VPN for early adopters (solid black line) and later adopters (dashed red line). The x-axis for Panel (b) is calendar time in weeks. Each vertical line in Panel (b) corresponds to an effective date for the early adopting states: Louisiana in January, 2023; Utah in May, 2023; Mississippi, Virginia, and Arkansas in July 2023, and Montana and North Carolina in January, 2024.



(a) Relative Week Time Series for Early Adopters



(b) Calendar-Week Time Series for Early versus Late Adopters

Figure 21: Search Interest in Porn

Notes: This figure shows the raw means for search interest in porn among the early adopters and later adopters in the sample period: January 2, 2022 to August 2, 2024. Panel (a) reports the weekly average search interest in porn for all early adopting states in my sample. The x-axis for Panel (a) is the week relative to the effective date of the law. I highlight the effective date of the law with a vertical line at week zero. Panel (b) reports the weekly average search interest in porn for early adopters (solid black line) and later adopters (dashed red line). The x-axis for Panel (b) is calendar time in weeks. Each vertical line in Panel (b) corresponds to an effective date for the early adopting states: Louisiana in January, 2023; Utah in May, 2023; Mississippi, Virginia, and Arkansas in July 2023, and Montana and North Carolina in January, 2024.

9.2 Weekly Event Studies

Figures 22-26 report the results from estimating the event study from Section 5 at the weekly frequency. In each graph, the point estimates for each β_j are reported alongside the 95% confidence intervals. The omitted period in each graph is the week before the law went into effect.

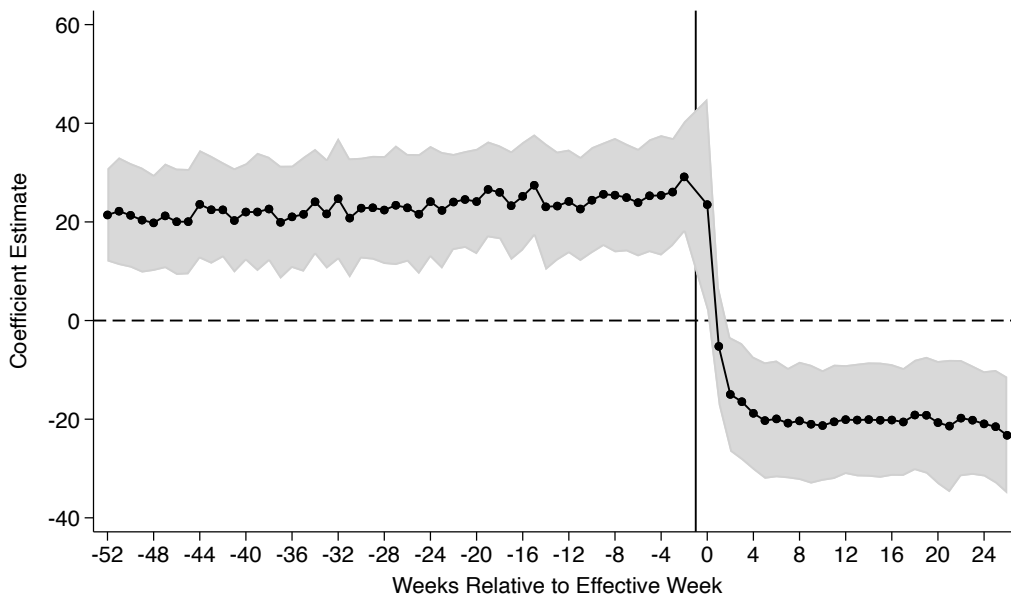


Figure 22: Search Interest in Pornhub

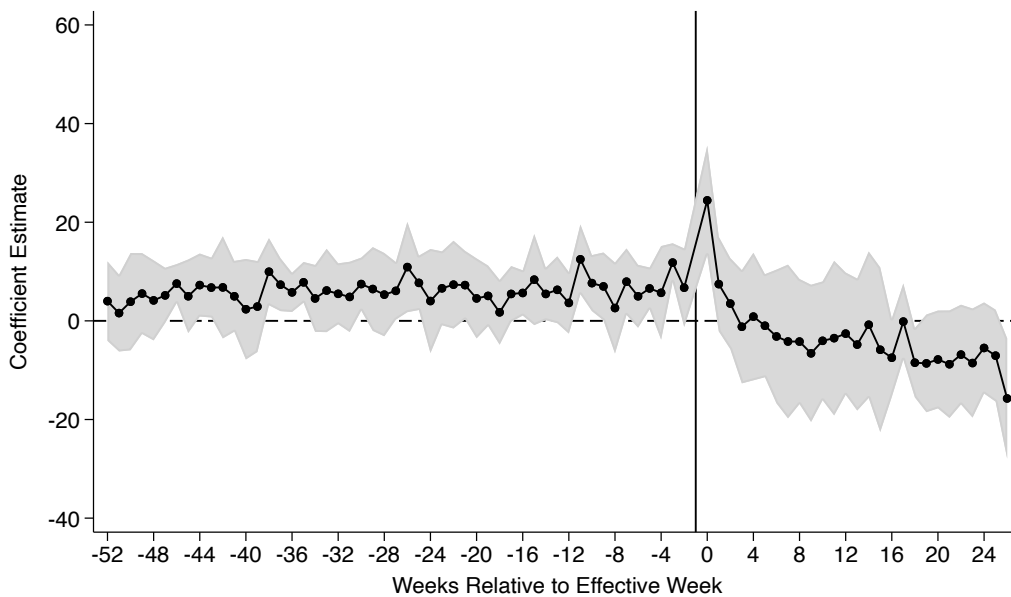


Figure 23: Search Interest in Xhamster

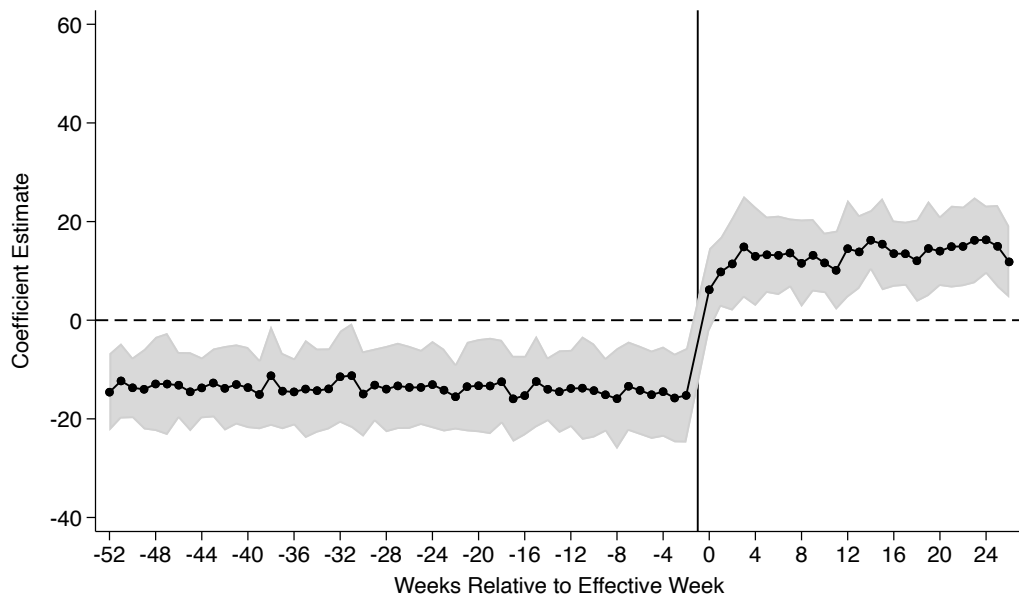


Figure 24: Search Interest in Xvideos

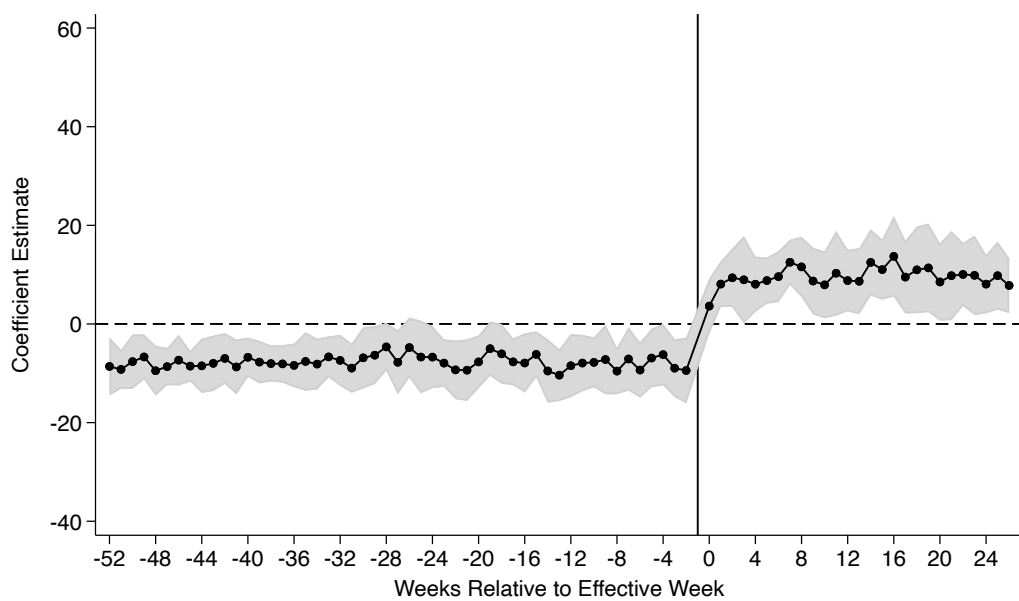


Figure 25: Search Interest in XNXX

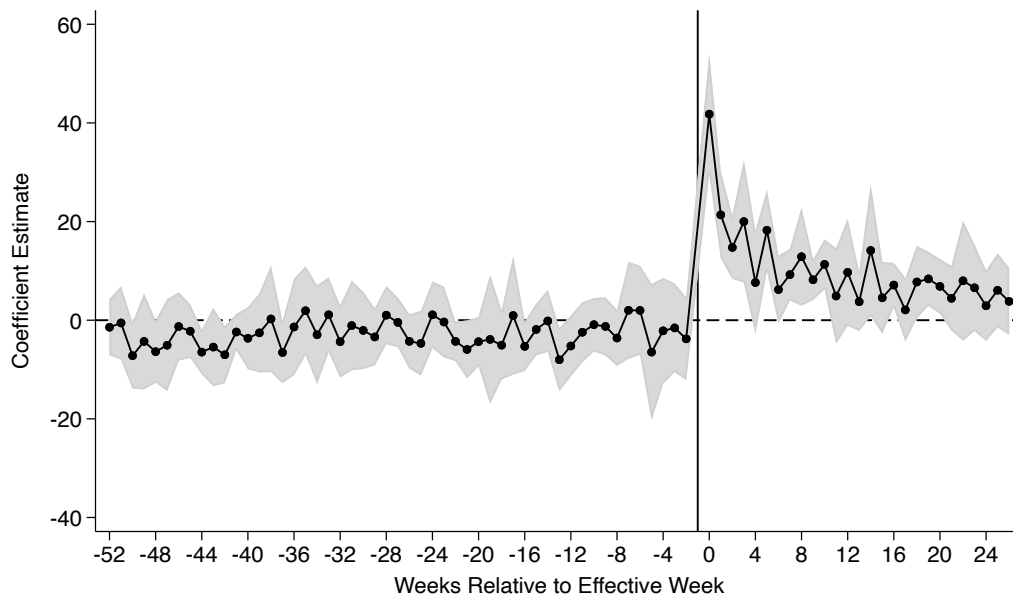


Figure 26: Search Interest in VPN

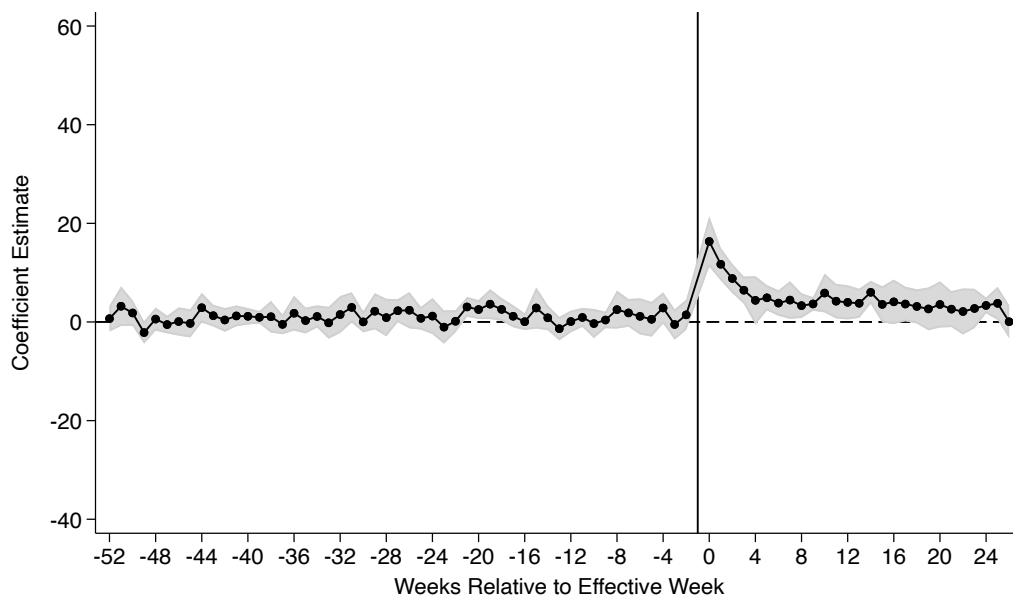


Figure 27: Search Interest in Porn

9.3 Robustness Using Different Reference Periods

The Figures in this section report the event studies from the main specification with robustness to different specifications of the reference period.

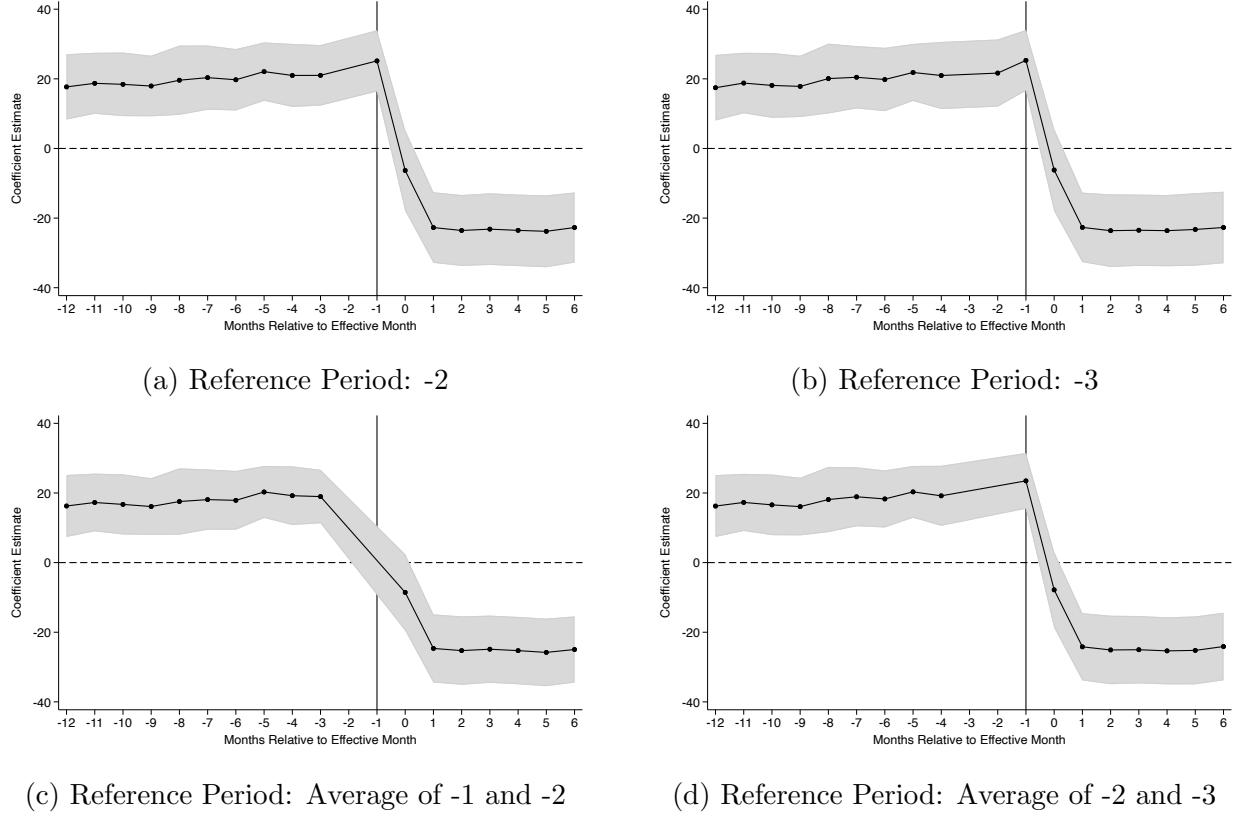
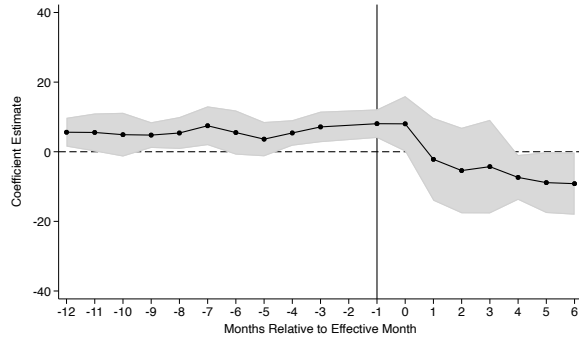
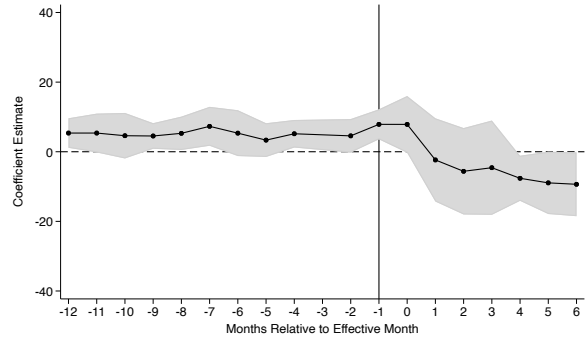


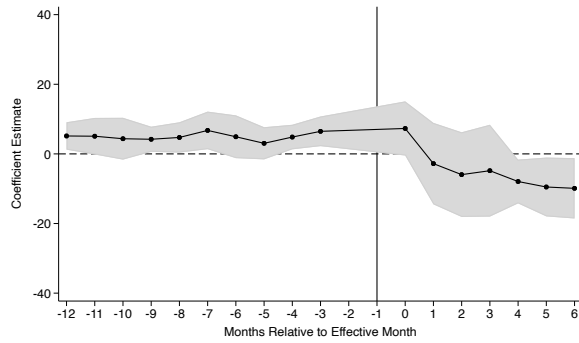
Figure 28: Effects on Search Interest in Pornhub



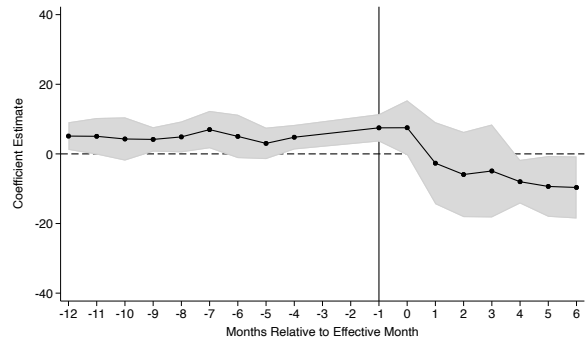
(a) Reference Period: -2



(b) Reference Period: -3

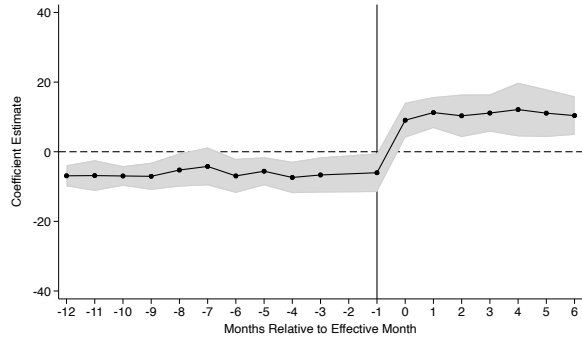


(c) Reference Period: Average of -1 and -2

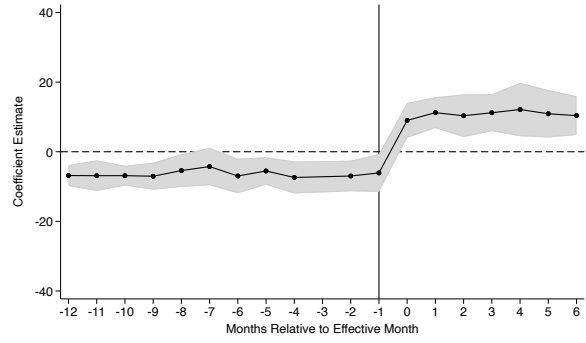


(d) Reference Period: Average of -2 and -3

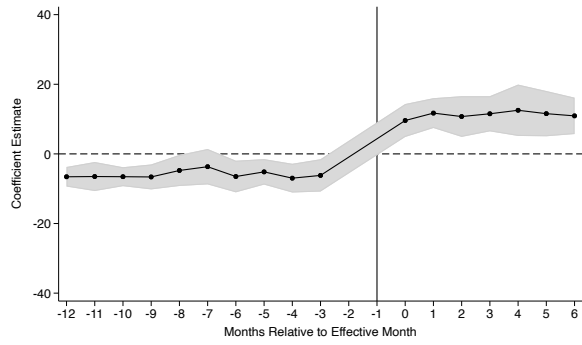
Figure 29: Effects on Search Interest in Xhamster



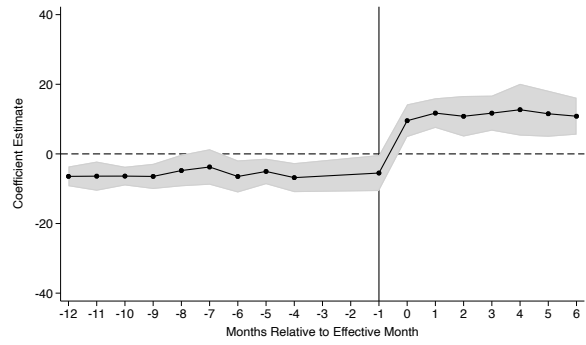
(a) Reference Period: -2



(b) Reference Period: -3

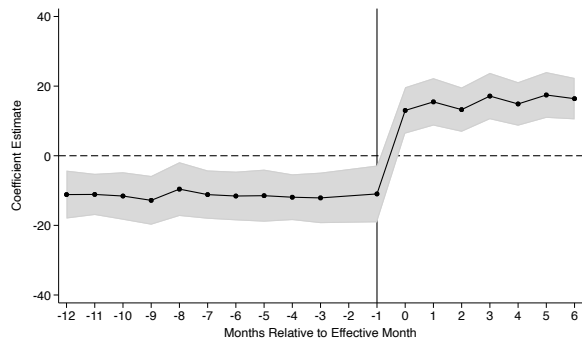


(c) Reference Period: Average of -1 and -2

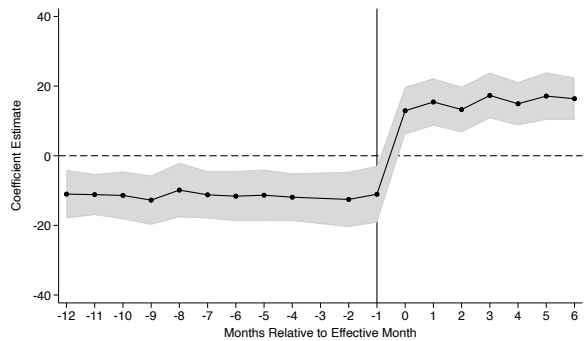


(d) Reference Period: Average of -2 and -3

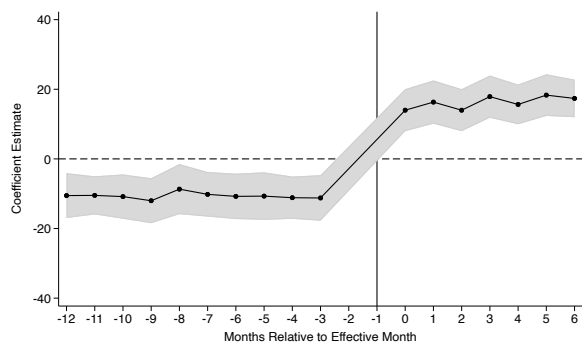
Figure 30: Effects on Search Interest in XNXX



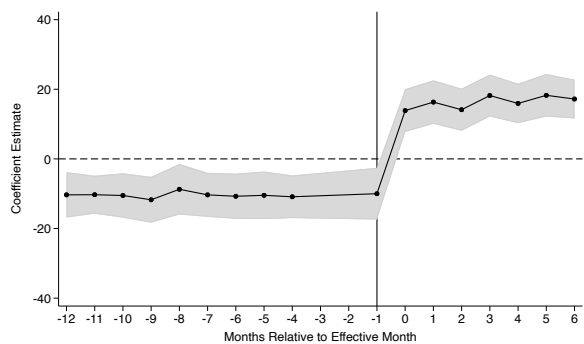
(a) Reference Period: -2



(b) Reference Period: -3

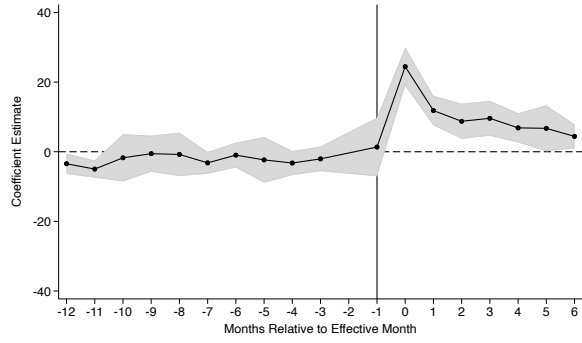


(c) Reference Period: Average of -1 and -2

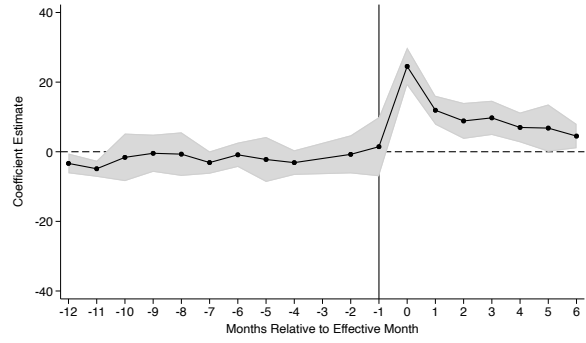


(d) Reference Period: Average of -2 and -3

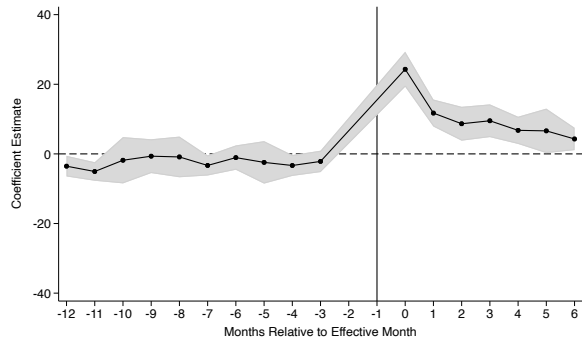
Figure 31: Effects on Search Interest in Xvideos



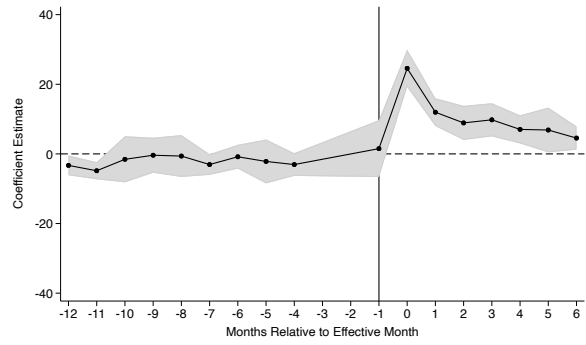
(a) Reference Period: -2



(b) Reference Period: -3

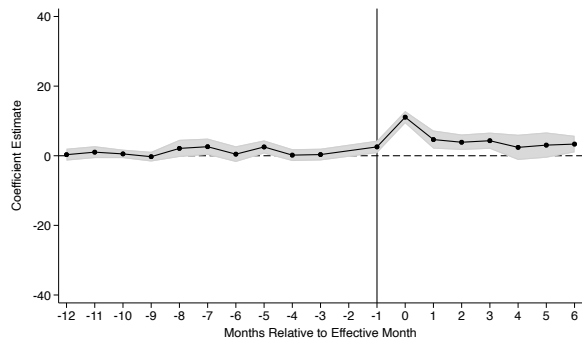


(c) Reference Period: Average of -1 and -2

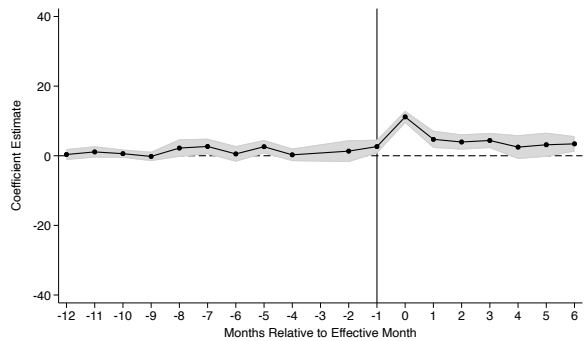


(d) Reference Period: Average of -2 and -3

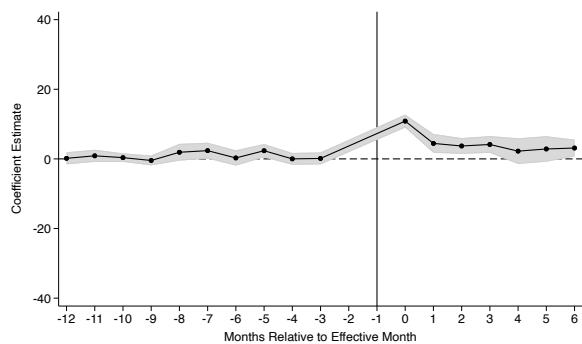
Figure 32: Effects on Search Interest in VPN



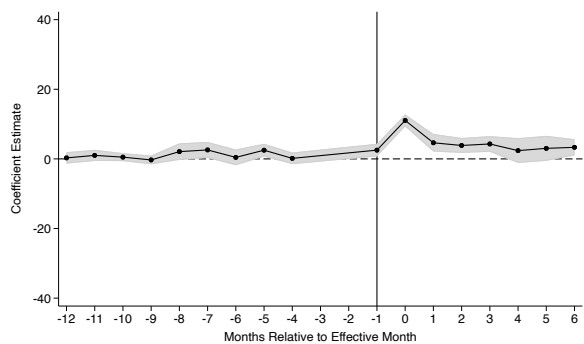
(a) Reference Period: -2



(b) Reference Period: -3



(c) Reference Period: Average of -1 and -2

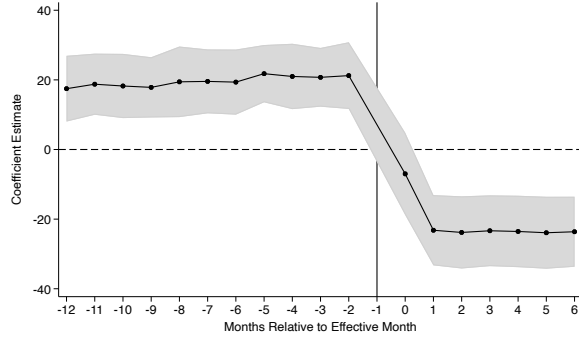


(d) Reference Period: Average of -2 and -3

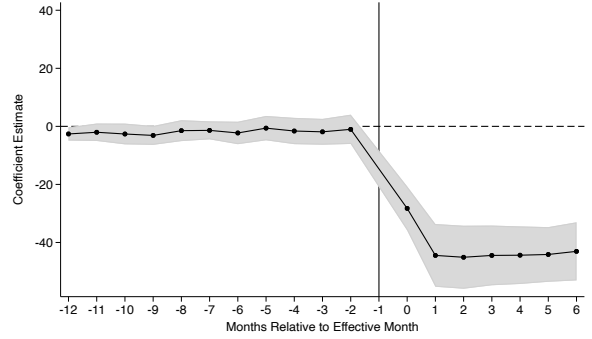
Figure 33: Effects on Search Interest in Porn

9.4 Comparison to Specification Without State Fixed Effects

In this section, I include, for each term, a side by side comparison of the results from the main specification with both state and calendar time fixed effects and a specification where I exclude state fixed effects. The specification without state fixed effects gives insight into how much of the results from the main specification are driven by the within-state demeaning.

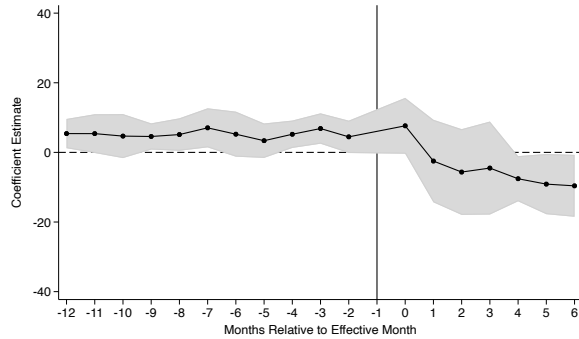


(a) Main Specification (calendar time and state fixed effects)

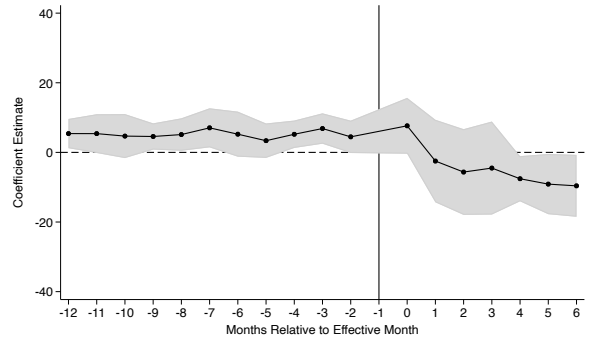


(b) Without State Fixed Effects

Figure 34: Effects on Search Interest in Pornhub

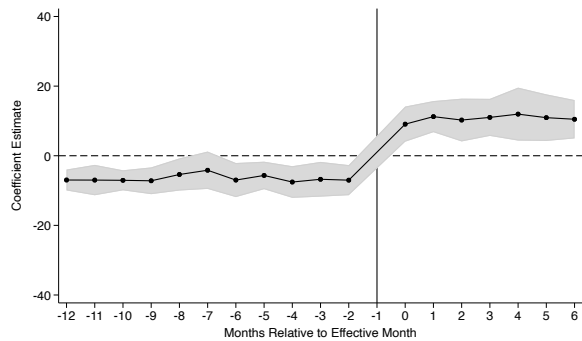


(a) Main Specification (calendar time and state fixed effects)

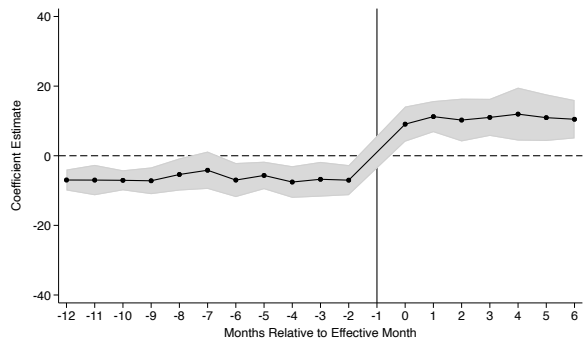


(b) Without State Fixed Effects

Figure 35: Effects on Search Interest in Xhamster

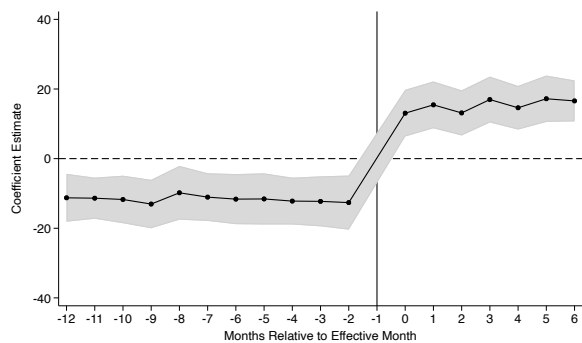


(a) Main Specification (calendar time and state fixed effects)

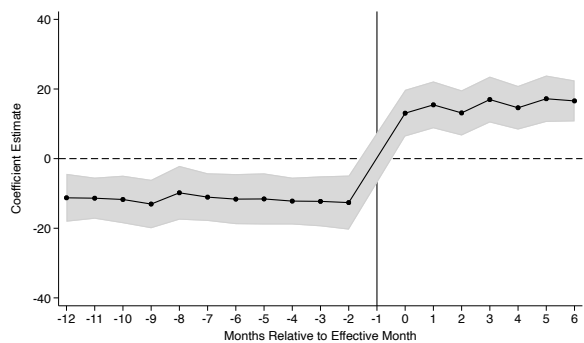


(b) Without State Fixed Effects

Figure 36: Effects on Search Interest in XNXX

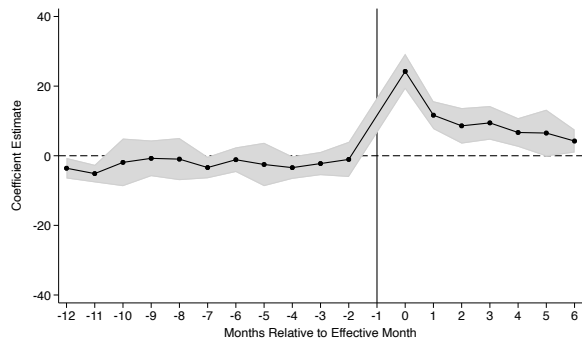


(a) Main Specification (calendar time and state fixed effects)

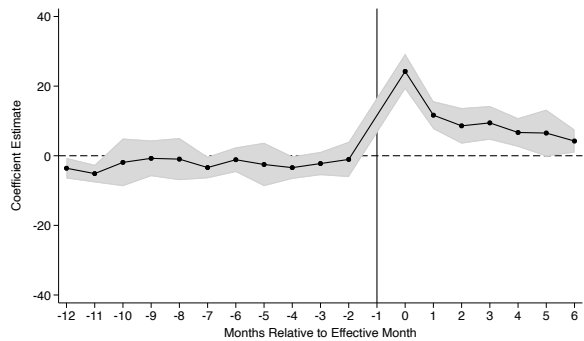


(b) Without State Fixed Effects

Figure 37: Effects on Search Interest in Xvideos

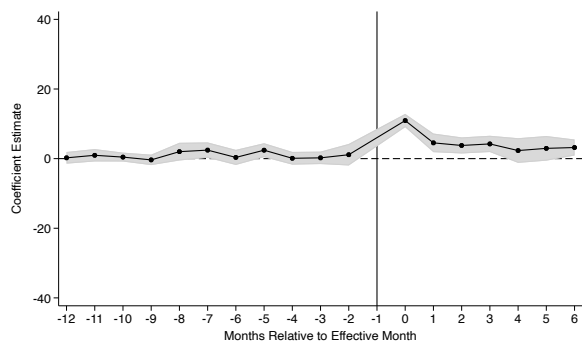


(a) Main Specification (calendar time and state fixed effects)

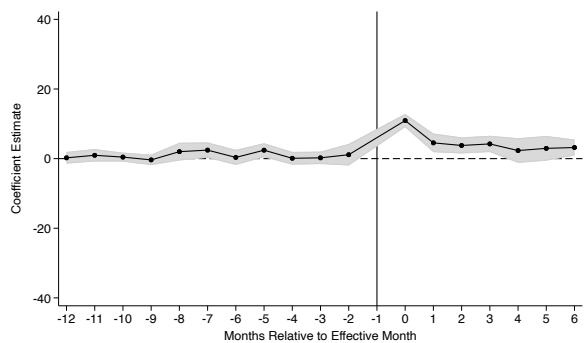


(b) Without State Fixed Effects

Figure 38: Effects on Search Interest in VPN



(a) Main Specification (calendar time and state fixed effects)



(b) Without State Fixed Effects

Figure 39: Effects on Search Interest in Porn

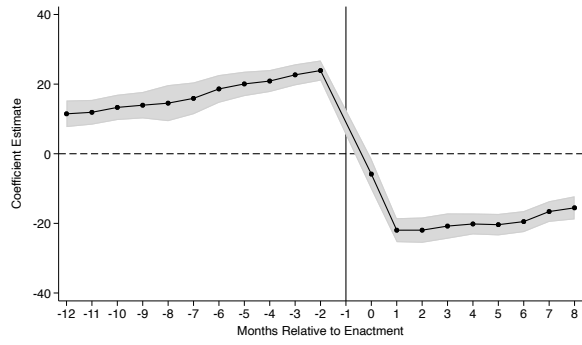
9.5 Alternative Specifications

9.5.1 Using Only Variation in Event Time

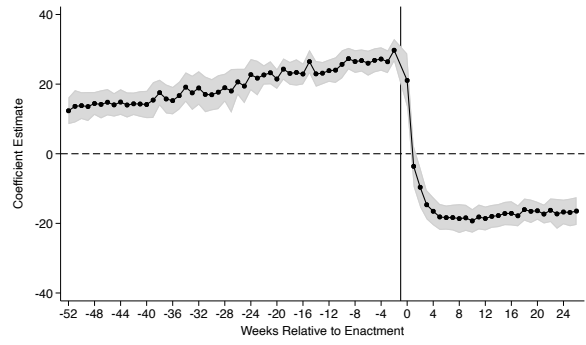
In this alternative specification, I use an event study that leverages the staggered enactment of age verification laws. I use only ever-treated states and use the variation in event-time, defined as the calendar period in which the law went into effect, to identify the effect of the policy. The intuition for this strategy is that the timing of the event is as good as random, so those treated later can serve as controls for those treated earlier.

$$y_{st} = \gamma_t + \sum_{j=-12}^{j=-2} \beta_j \times D_{s,t-j} + \sum_{j=0}^{j=8} \beta_j \times D_{s,t-j} + \varepsilon_{st}, \quad (1)$$

where y_{st} represents the search interest for a term in state s in time t . The specification includes calendar time (γ_t) fixed effects. $D_{s,t-j}$ is an indicator for event time, j , meaning that the effective date for a law in state s is j periods before this observation's calendar time, t . I split the summation into two terms representing the lags and leads of the event time, respectively. This split also makes explicit that the reference period is period $j = -1$, the month before the law takes effect. The coefficients of interest are the β_j s for $j \geq 0$, which capture the dynamic effects of treatment: the change in search interest for a given term relative to the omitted period, β_{-1} . I cluster the standard errors at the state level. I present the estimates of each β_j along with 95% confidence intervals below. The monthly and weekly estimates for each term are presented next to each other.

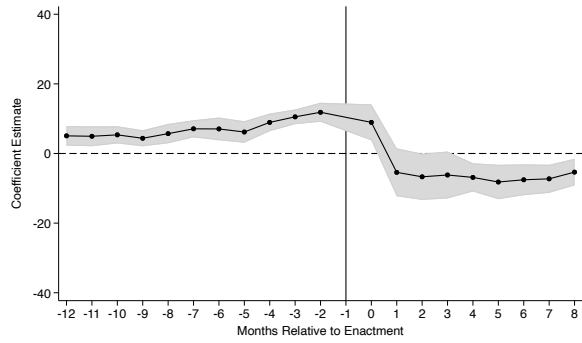


(a) Monthly

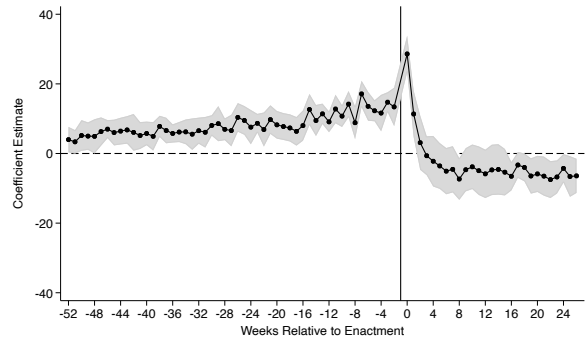


(b) Weekly

Figure 40: Search Interest in Pornhub

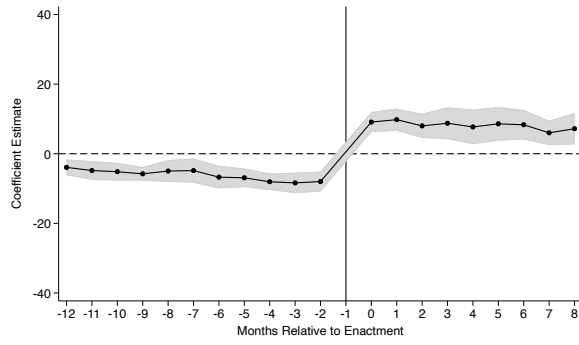


(a) Monthly

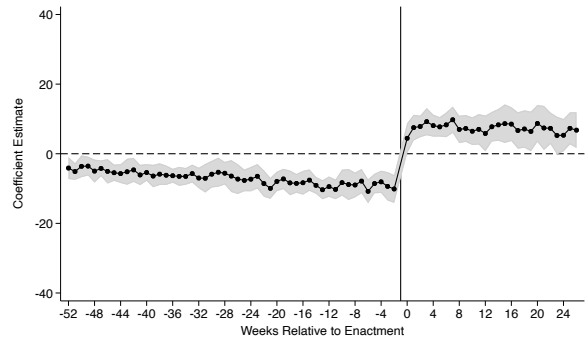


(b) Weekly

Figure 41: Search Interest in Xhamster

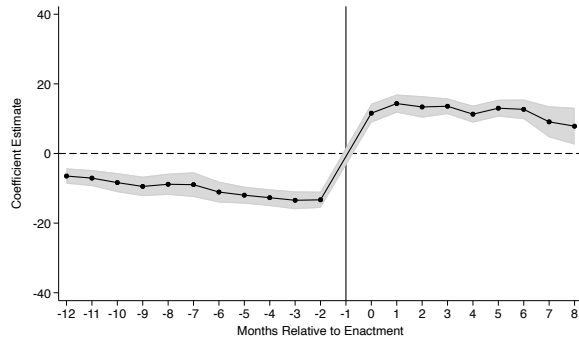


(a) Monthly

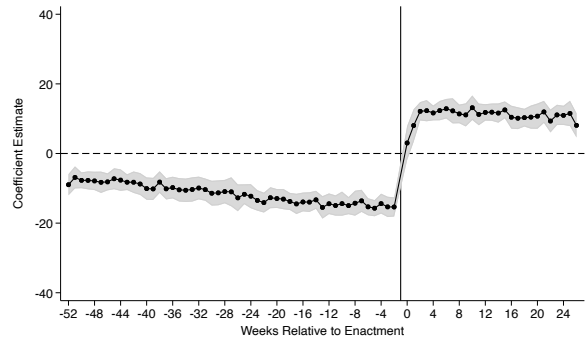


(b) Weekly

Figure 42: Search Interest in XNXX

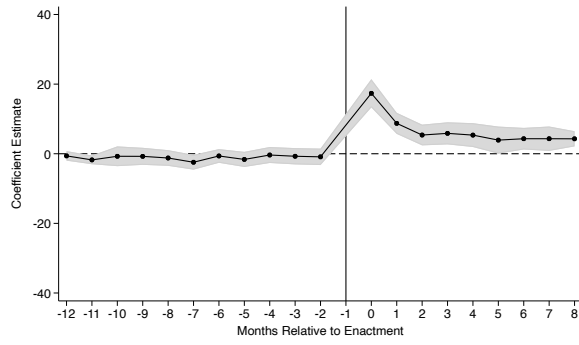


(a) Monthly

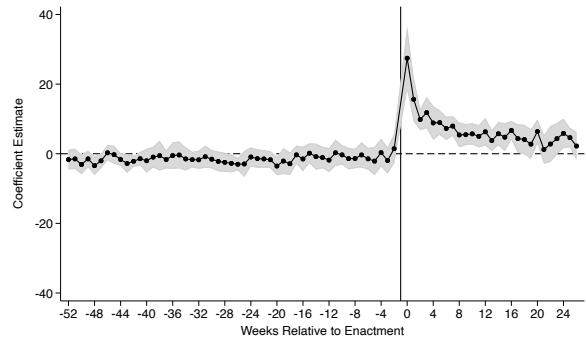


(b) Weekly

Figure 43: Search Interest in Xvideos

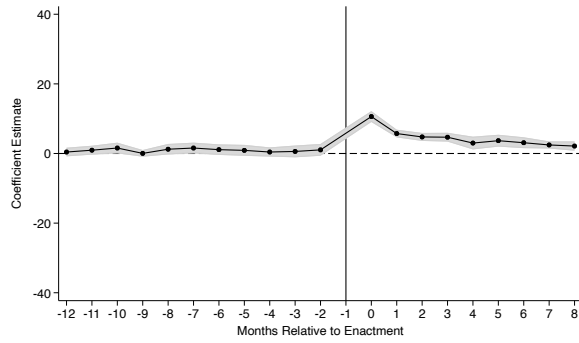


(a) Monthly

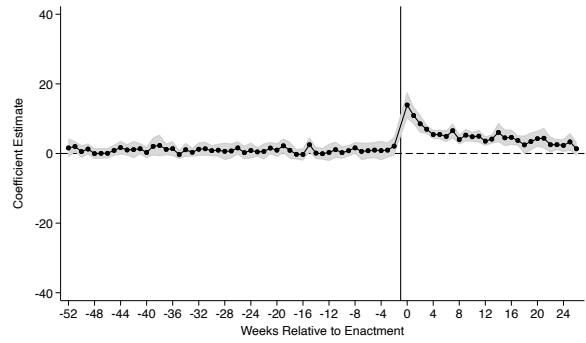


(b) Weekly

Figure 44: Search Interest in VPN



(a) Monthly



(b) Weekly

Figure 45: Search Interest in Porn

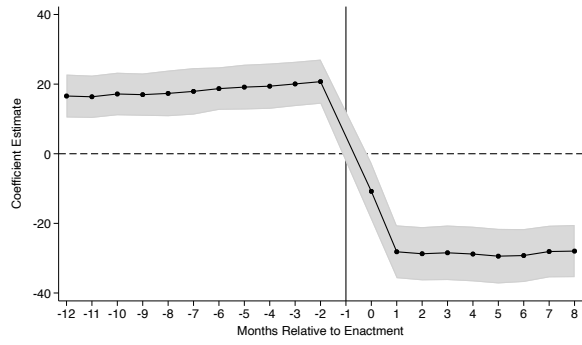
9.5.2 Comparing Adopting States to Non-Adopting States

In this alternative specification, I use a difference-in-differences framework that compares outcomes in states that adopt age verification laws relative to states that do not adopt before and after the laws' effective dates.²⁰ I rely on an event study framework to capture the dynamic effects of treatment. I estimate the event study for each search term separately. The event study for the monthly specification is:

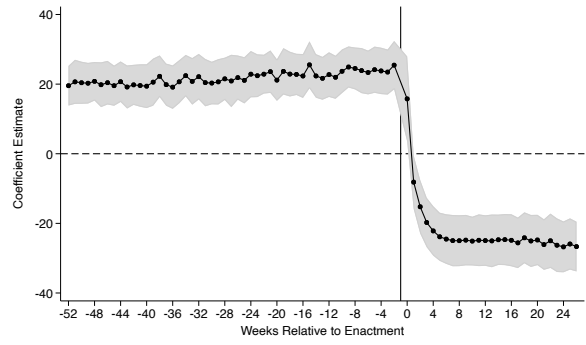
$$y_{st} = \alpha_s + \gamma_t + \sum_{j=-12}^{j=-2} \beta_j \times 1(\text{Adopter})_s \times D_{s,t-j} + \sum_{j=0}^{j=8} \beta_j \times 1(\text{Adopter})_s \times D_{s,t-j} + \varepsilon_{st}, \quad (2)$$

where y_{st} represents the search interest for a term in state s in time t . The specification includes state (α_s) and calendar time (γ_t) fixed effects. $1(\text{Adopter})_s$ is an indicator equal to one if a state adopts an age verification law by January 1, 2025. $D_{s,t-j}$ is an indicator for event time, j , meaning that the effective date for a law in state s is j periods before this observation's calendar time, t . The non-adopting states have $D_{s,t-j} = 0$ for every period, t . I split the summation into two terms representing the lags and leads of the event time, respectively. This split also makes explicit that the reference period is period $j = -1$, the month before the law takes effect. The coefficients of interest are the β_j s for $j \geq 0$, which capture the dynamic effects of treatment: the differential change in a term's search interest for adopting versus non-adopting states at a given period, j , relative to the omitted period, β_{-1} . I cluster the standard errors at the state level. I present the estimates of each β_j along with 95% confidence intervals below. The monthly and weekly estimates for each term are presented next to each other.

²⁰The adopters in my estimation sample are Louisiana, Utah, Mississippi, Virginia, Arkansas, Montana, North Carolina, Idaho, Kansas, Kentucky, Nebraska, Indiana, Alabama, Oklahoma, Florida, South Carolina, and Tennessee. I exclude Texas because it is unclear when the law really took effect due to the legal complications. I also exclude Georgia, Wyoming, South Dakota, North Dakota, Arizona, Ohio, and Missouri because these states have fewer than six months of post-treatment data.

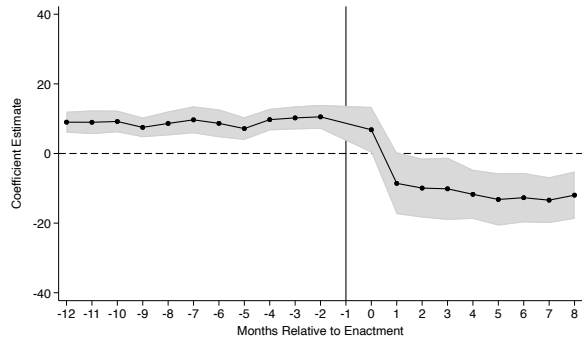


(a) Monthly

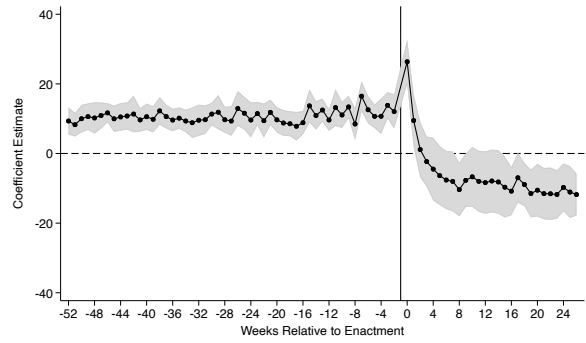


(b) Weekly

Figure 46: Search Interest in Pornhub

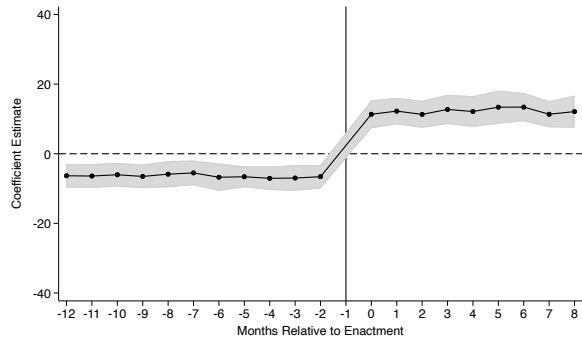


(a) Monthly

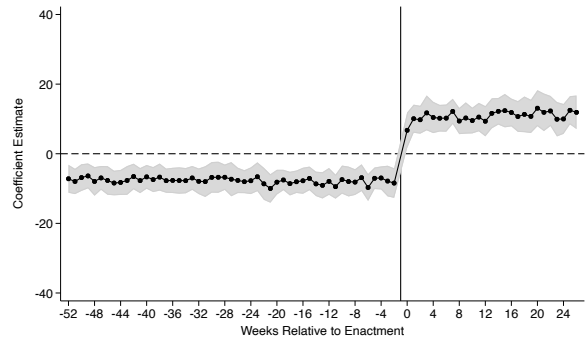


(b) Weekly

Figure 47: Search Interest in Xhamster

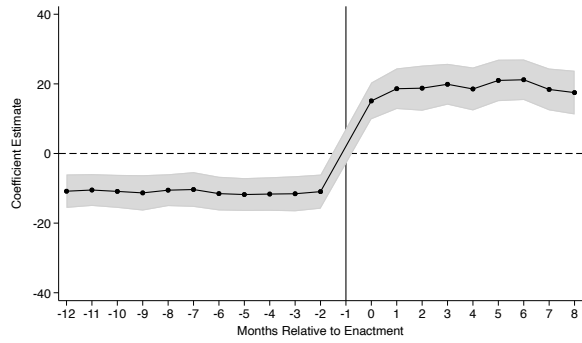


(a) Monthly

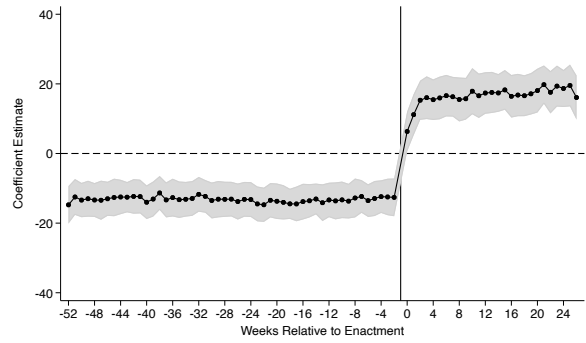


(b) Weekly

Figure 48: Search Interest in XNXX

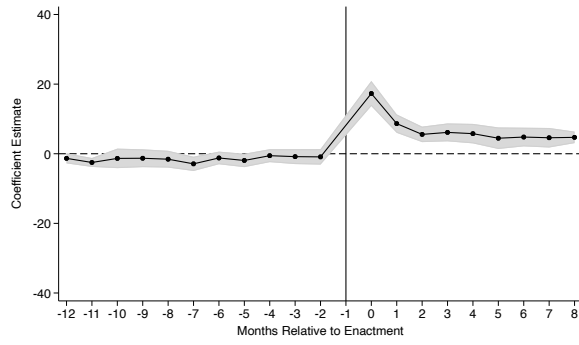


(a) Monthly

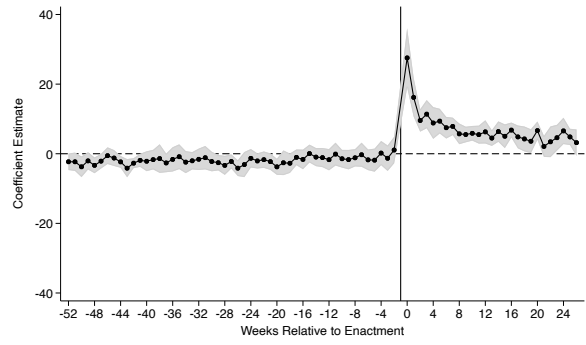


(b) Weekly

Figure 49: Search Interest in Xvideos

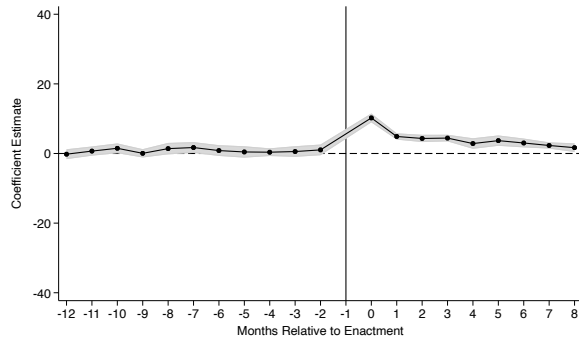


(a) Monthly

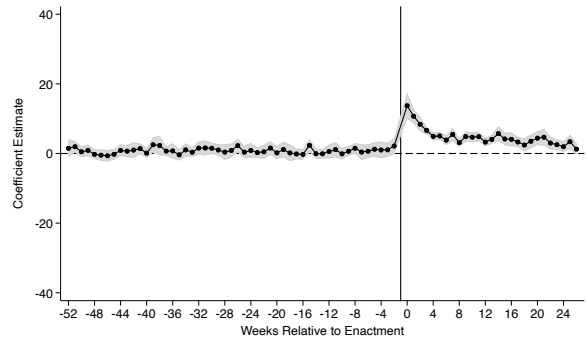


(b) Weekly

Figure 50: Search Interest in VPN



(a) Monthly



(b) Weekly

Figure 51: Search Interest in Porn